ĐẠI HỌC KHOA HỌC & CÔNG NGHỆ HÀ NỘI
VIETNAM – FRANCE UNIVERSITY
Tòa nhà Đào tạo và Dịch vụ
18 Hoàng Quốc Việt, Cầu Giấy, Hà Nội
Website: http://www.usth.edu.vn/

Genetic

Academic year: 1st year

Credit: 3 ECTS

Lecturer: NGUYEN Thy Ngoc

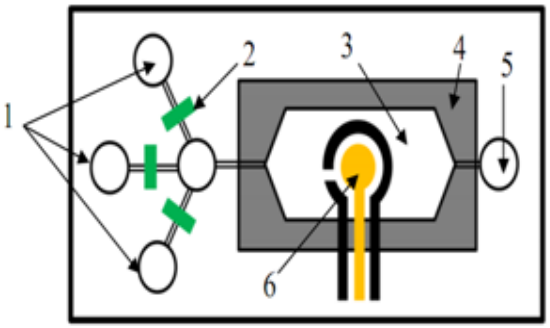
Email: nguyen-thy.ngoc@usth.edu.vn

Course description

Purpose of the course
This course discusses the principles of genetics with application to the study of biological functions. The topics include: structure and function of genes, chromosomes and genomes, genetic variations, population genetics, use of genetic methods to analyze protein function, gene regulation and inherited disease.

Time schedule	
Lecture	20 hours
Exercises	10 hours

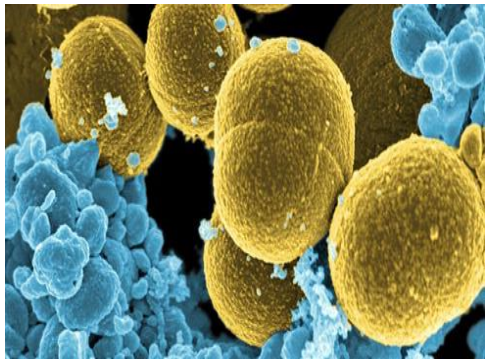
Why we have to study “Genetic”?



NANO



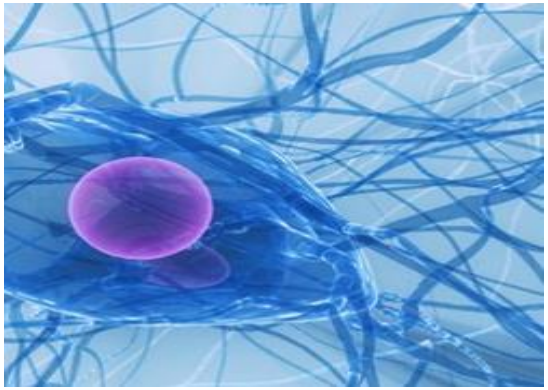
ICT



WEO



ENERGY



MST



FST

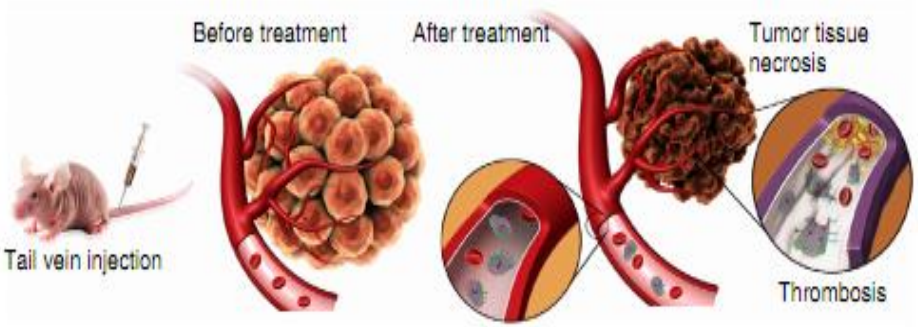
Nanorobotics

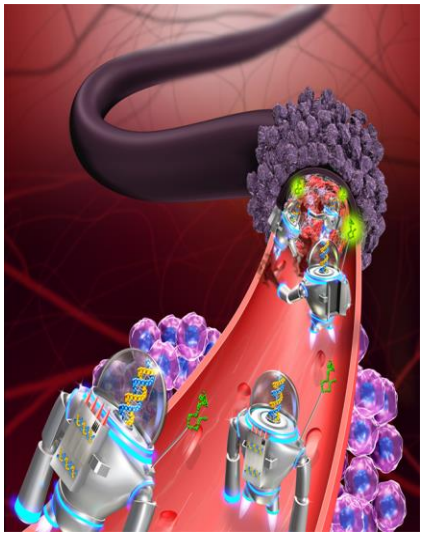
nature biotechnology

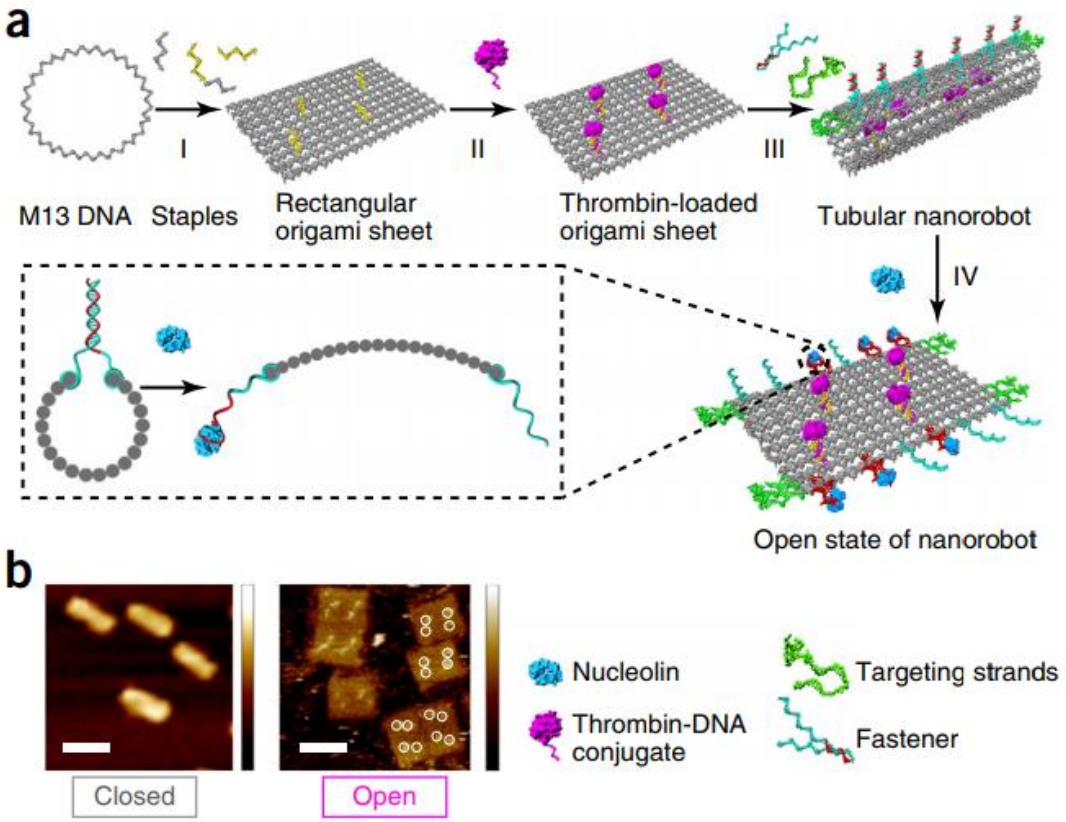
2018

A DNA nanorobot functions as a cancer therapeutic in response to a molecular trigger *in vivo*

Suping Li[1,2,10], Qiao Jiang[1,10], Shaoli Liu[1,2,10], Yinlong Zhang[1,3,10], Yanhua Tian[1,4], Chen Song[1], Jing Wang[1], Yiguo Zou[1], Gregory J Anderson[5], Jing-Yan Han[6], Yung Chang[7], Yan Liu[7], Chen Zhang[8], Liang Chen[9], Guangbiao Zhou[8], Guangjun Nie[1,2], Hao Yan[7], Baoquan Ding[1,2] & Yuliang Zhao[1,2]







Course contents

- Chapter I: DNA (Deoxyribonucleic acid), RNA.
- Chapter II: DNA replication.
- Chapter III: Transcription – Translation - Protein. (E)
- Chapter IV: Genetic variation – genetic markers.
- Chapter V: Mendel’s law & Morgan’s law. (E)
- Chapter VI: Linkage disequilibrium, Hardy Weinberg equilibrium. (E)
- Chapter VII: Introduction to Epigenetics.

Chapter I: DNA, RNA

Genetics: what is it?

- What is genetics?

"Genetics is the study of **heredity**, the process in which a parent passes certain **genes** onto their children."



- What does that mean?

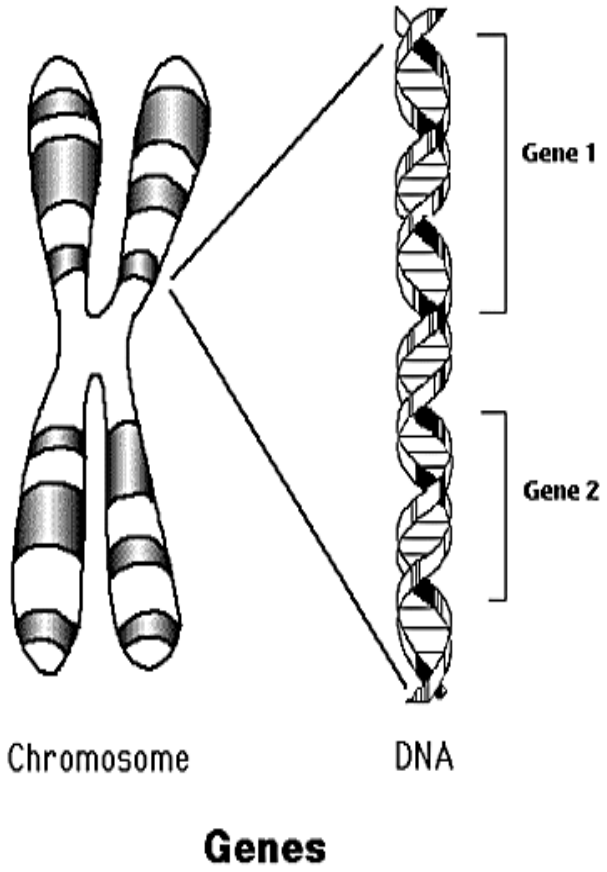
Children **inherit** their biological parents' genes that express specific **traits**, such as some physical characteristics, natural talents, and genetic disorders.

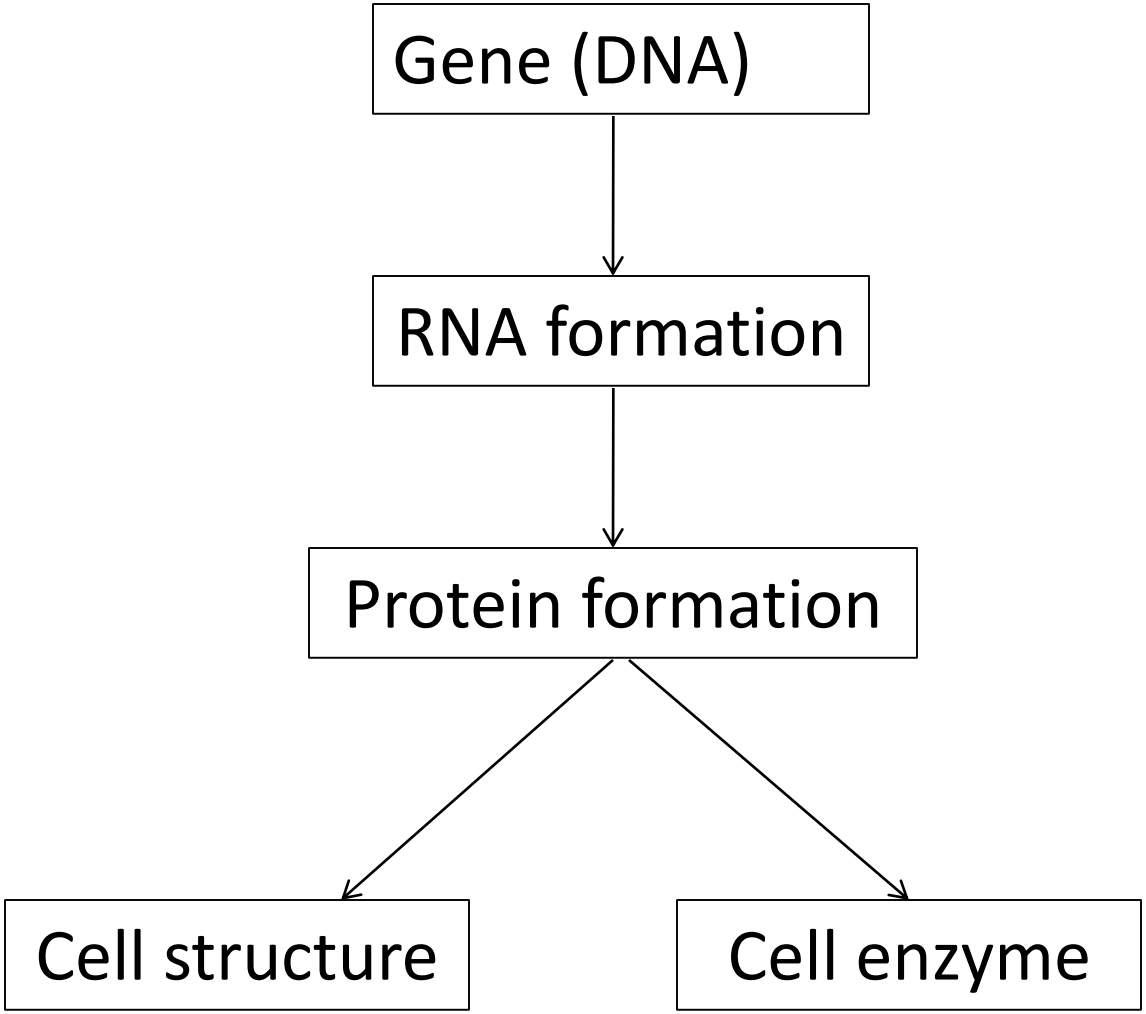
Watch video



Genetic Terminology

- **Gene** – basic unit of genetic information. Genes determine the inherited characters.
- **Genome** – the collection of genetic information.
- **Chromosomes** – storage units of *genes*.
- **DNA** - is a nucleic acid that contains the genetic instructions specifying the biological development of all cellular forms of life



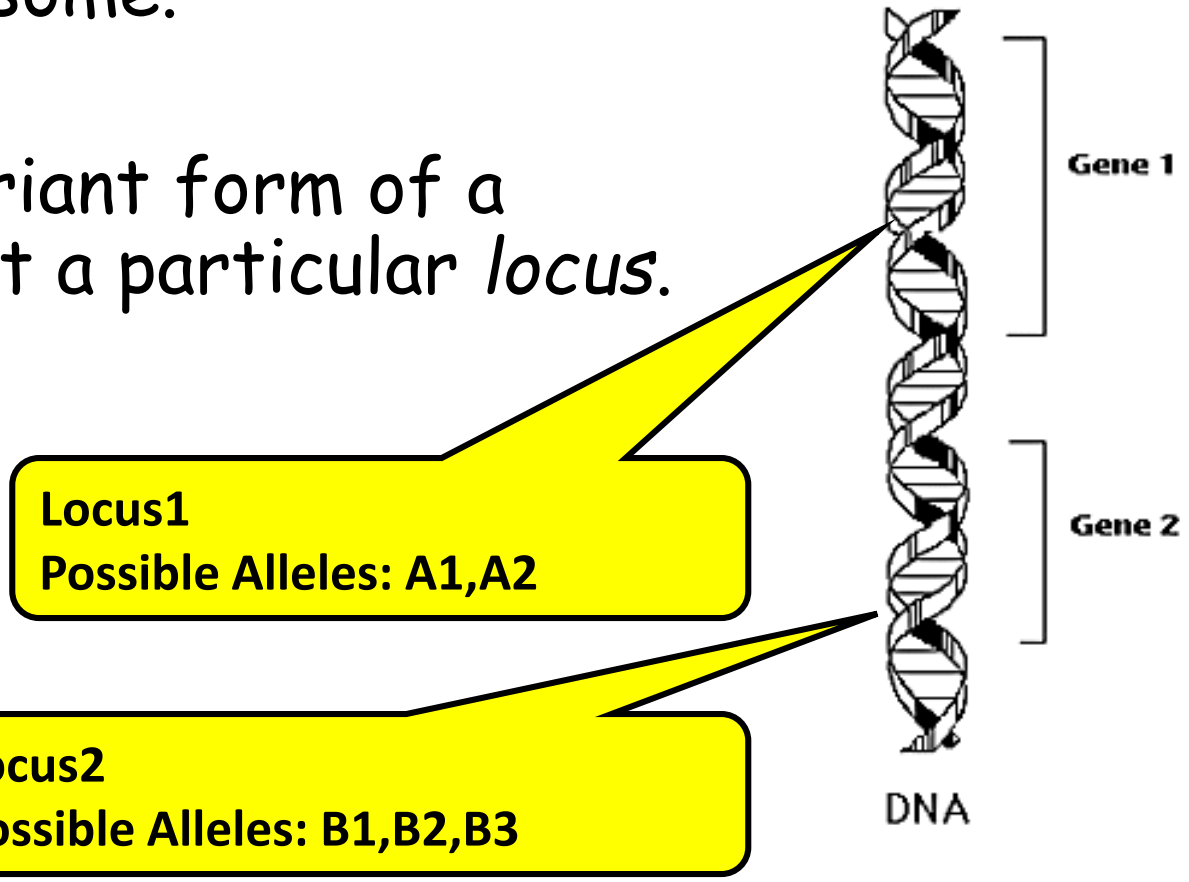

Gene (DNA)
RNA formation
Protein formation
Cell structure
Cell enzyme

Watch clip



Chromosome Logical Structure

- **Locus** – location of a *gene/marker* on the chromosome.
- **Allele** – one variant form of a gene/marker at a particular *locus*.



Genetic Concepts

- **Heredity** describes how some traits are passed from parents to their children.
- The traits are expressed by **genes**, which are small sections of DNA that are coded for specific traits.
- Genes are found on **chromosomes**.
- Humans have two sets of **23** chromosomes—one set from each parent.

Timeline of Experiments

1871 – Miescher – identifies Nucleic Acid

1928 – Griffith – “transformation” of bacteria

1944 – Avery, MacLeod and McCarty – DNase expt

1950 – Chargaff – G&C, A&T

1952 – Hershey and Chase – Blender expt

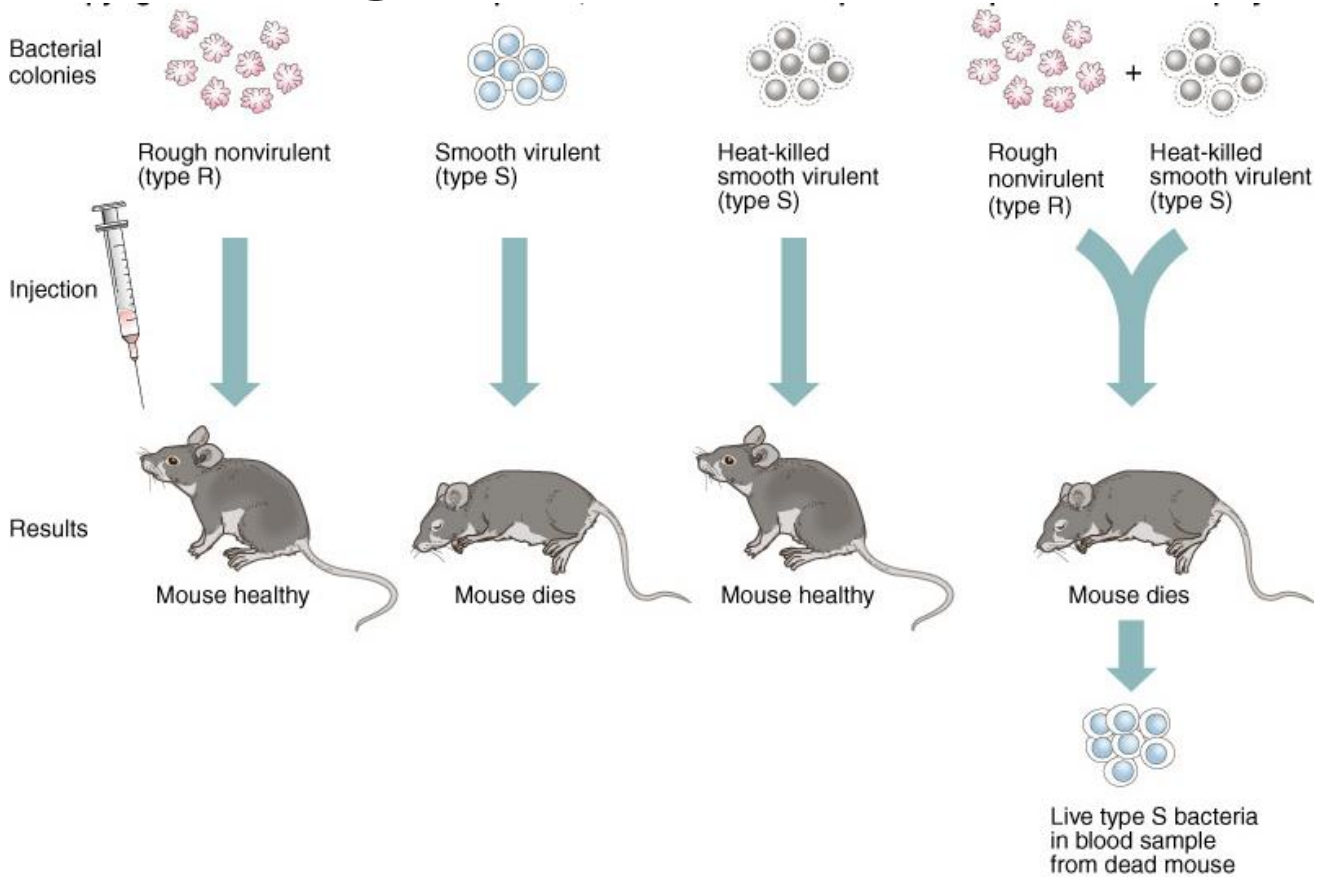
1953 – Franklin – picture of DNA

1953 – Watson and Crick – Double Helix structure and base pairing

Frederick Griffith

1928

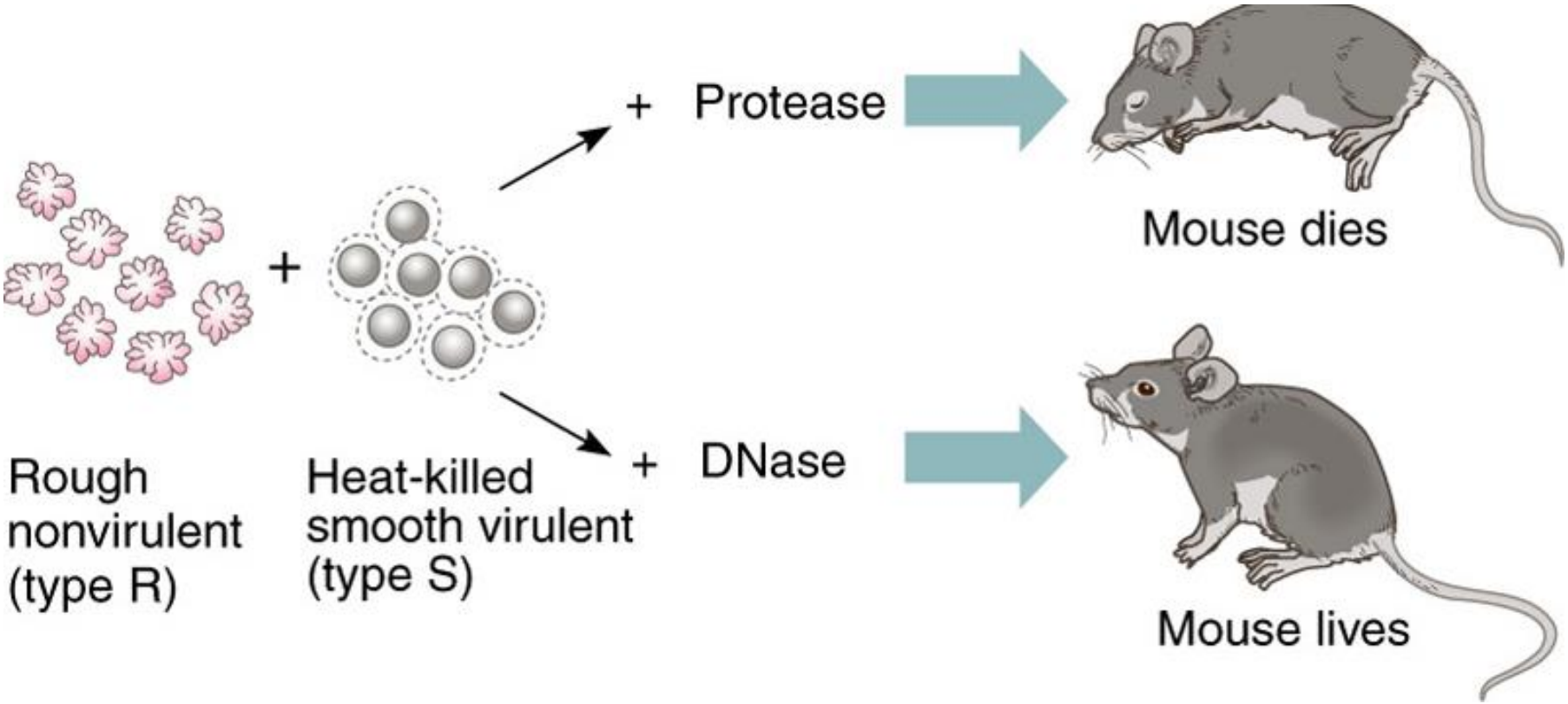
- Transforming factor in bacteria



Avery, MacLeod, McCarty

1944

- DNase vs. Protease

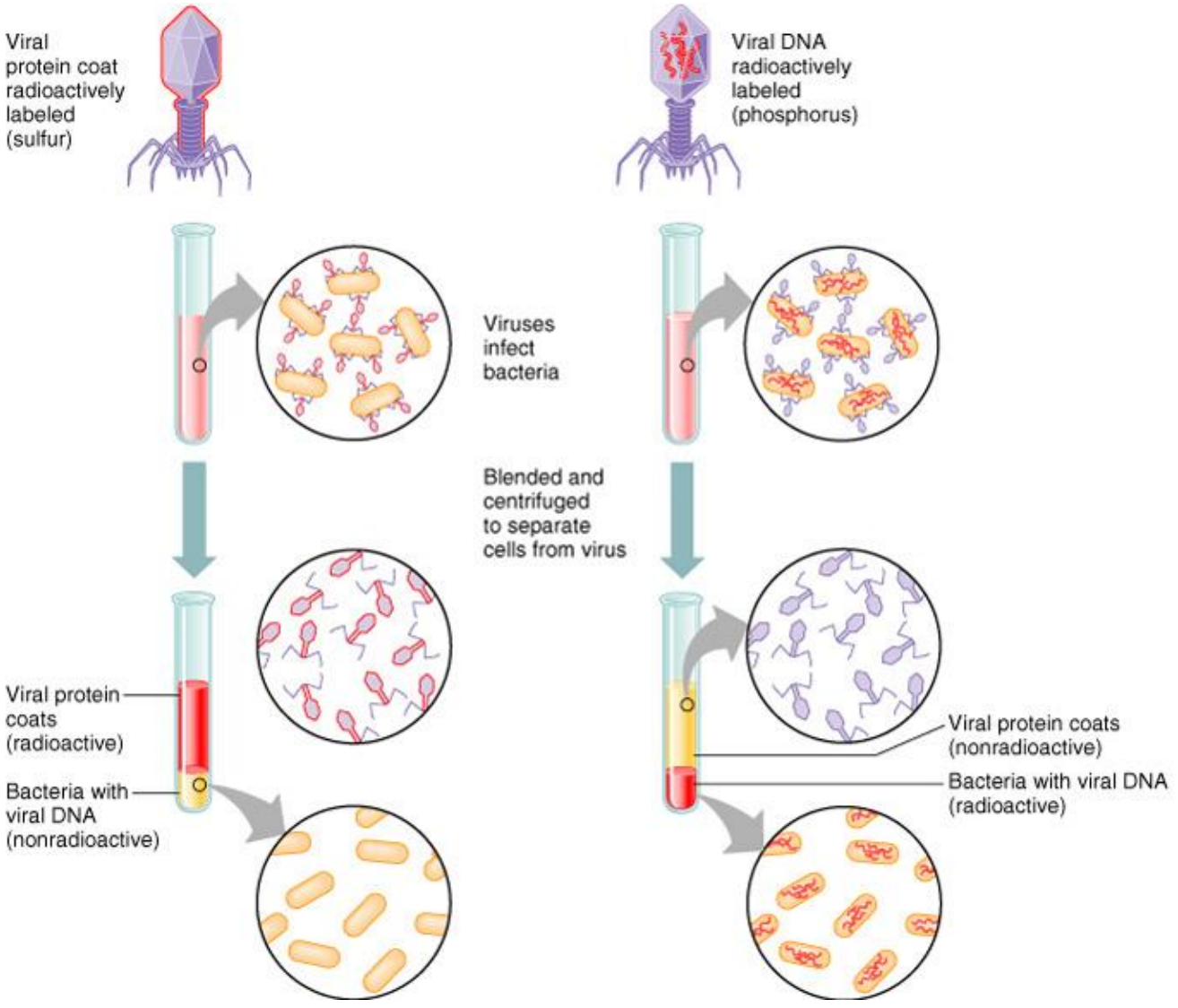


Hershey and Chase

1952

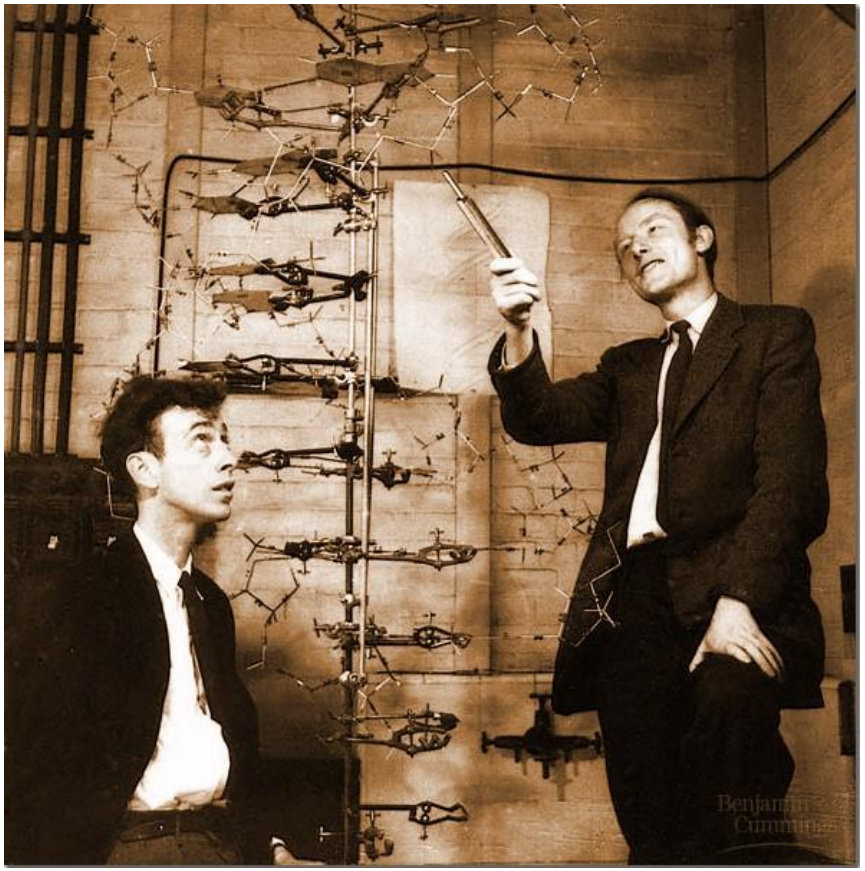
- Blender

Experiment



Discovering the structure of DNA

- Structure was discovered in 1953 by James Watson and Francis Crick



Watson and Crick

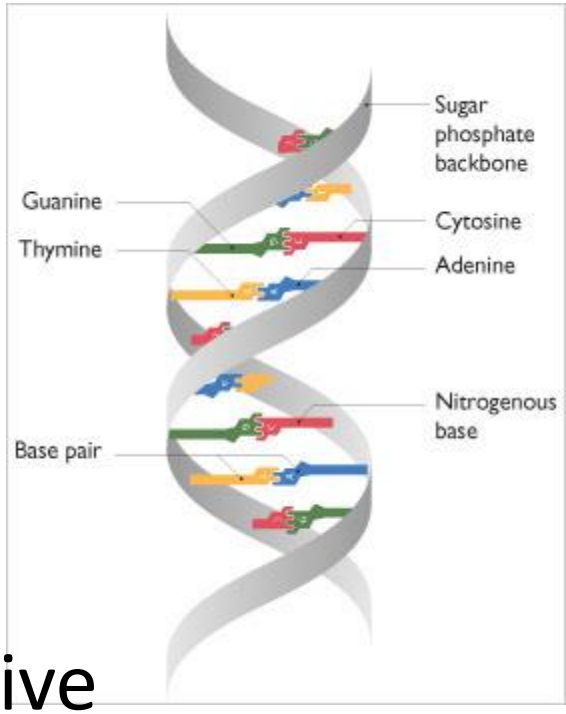
April 25th, 1953

- Seen Franklin's picture:
  - Assumed Sugar-Phosphate backbone
- Knew Chargaff's rule:
  - A&T, G&C must bind each other
- Determined turning radius of beta-Helix
- Realized it must be two complementary strands because of base pairing
- Determined it was a double Helix

2005

We know:

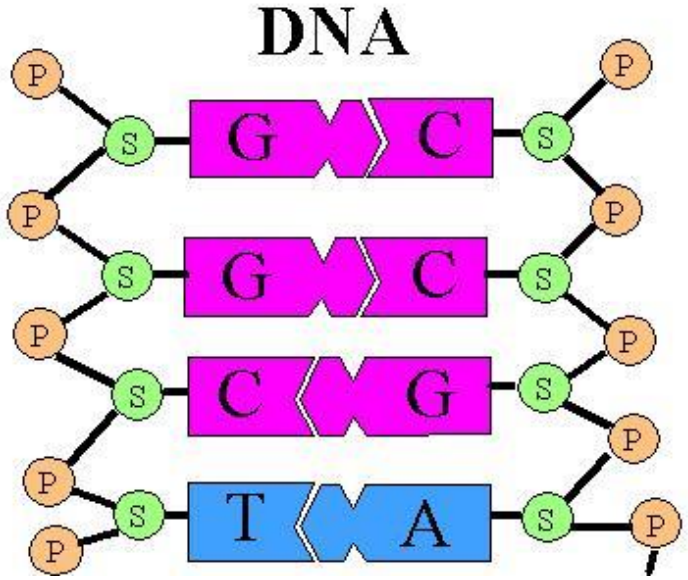
- DNA is the hereditary material
- DNA has a double helix structure
  - Made of four bases; A,T,C,G
  - Sugar-Phosphate backbone
- DNA replication is semi-conservative
- How to do PCR



Discovering the structure of DNA

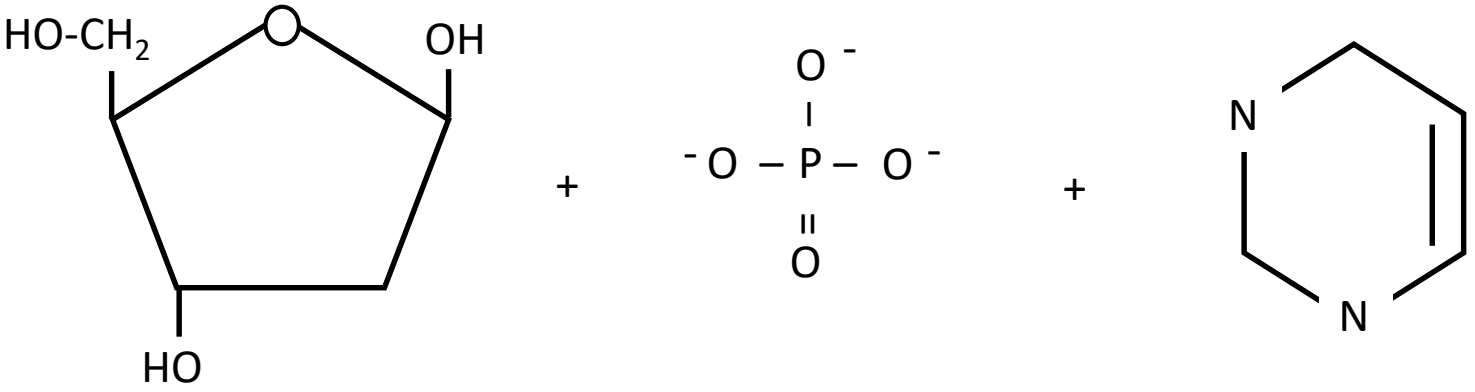
- DNA = **D**eoxyribose **n**ucleic **a**cid

- Made out of sugars (deoxyribose), phosphates and nitrogen bases



What is DNA?

- Deoxyribonucleic Acid:
  - String of nucleotides
    - Nucleotides made up of three parts:

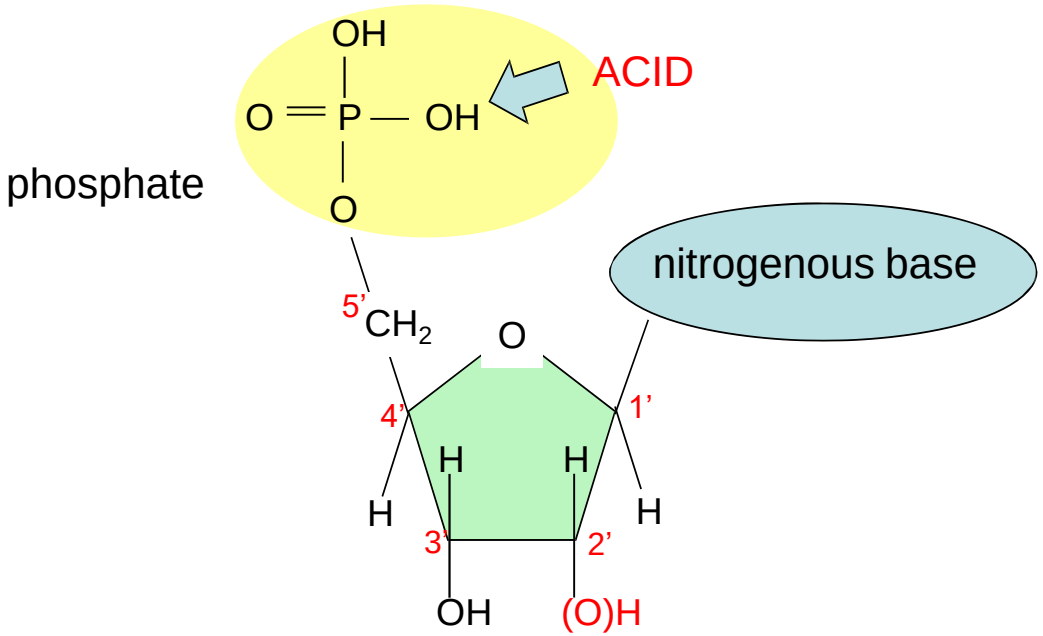


deoxyribose (a sugar)

phosphate

cyclic amine (base)

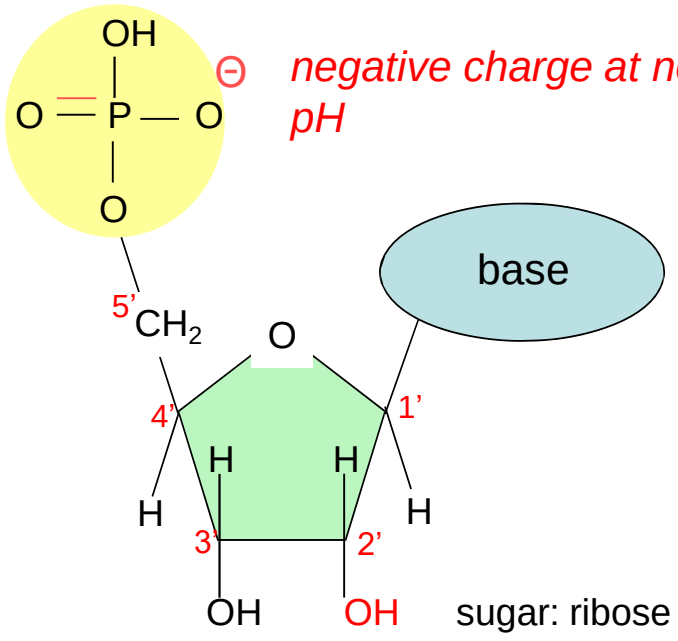
Nucleic acids



Nucleic acids

mainly located in cell nuclei
polymers of nucleotides (up to 100 000 000 units !!) water soluble

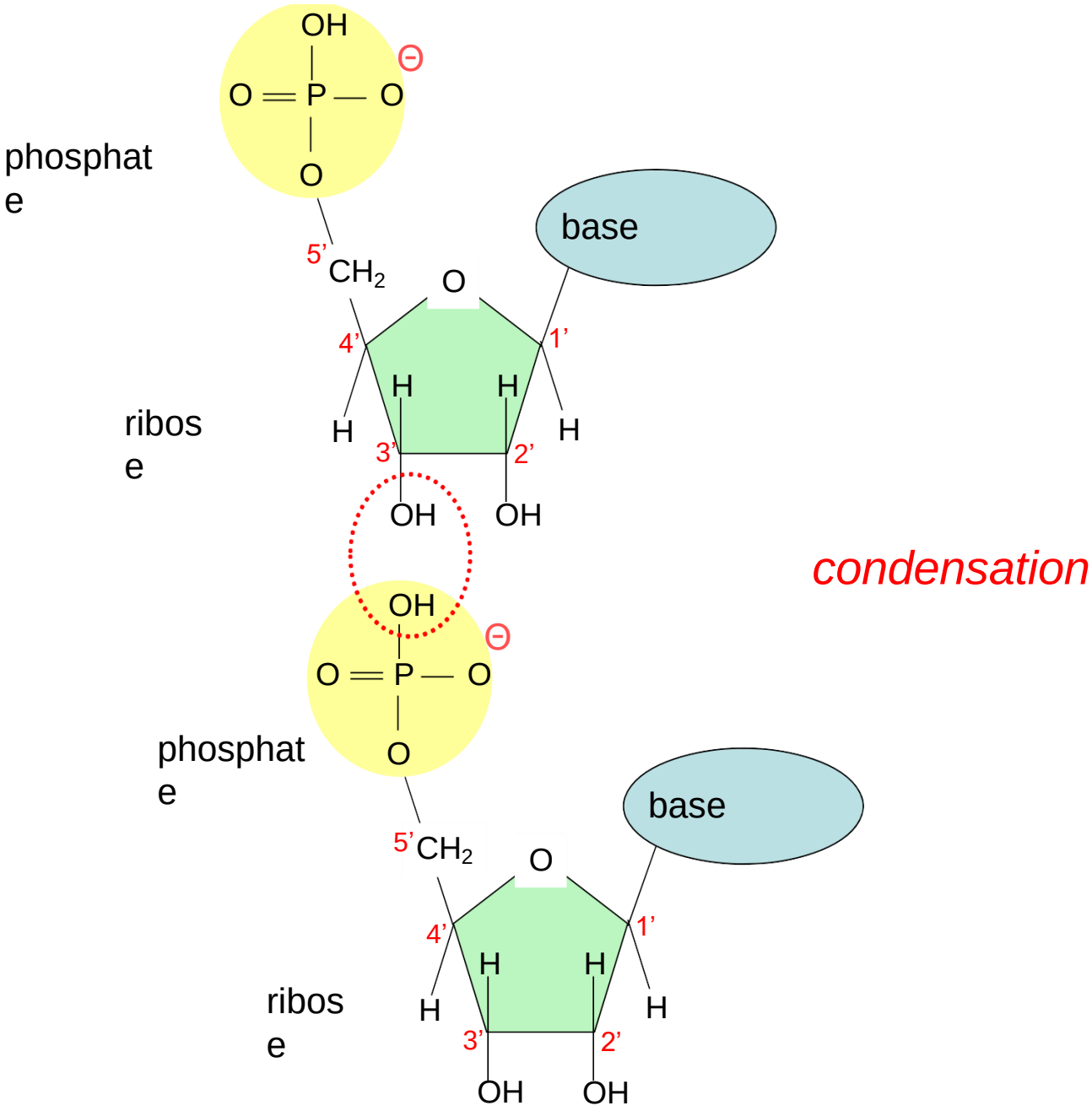
phosphate

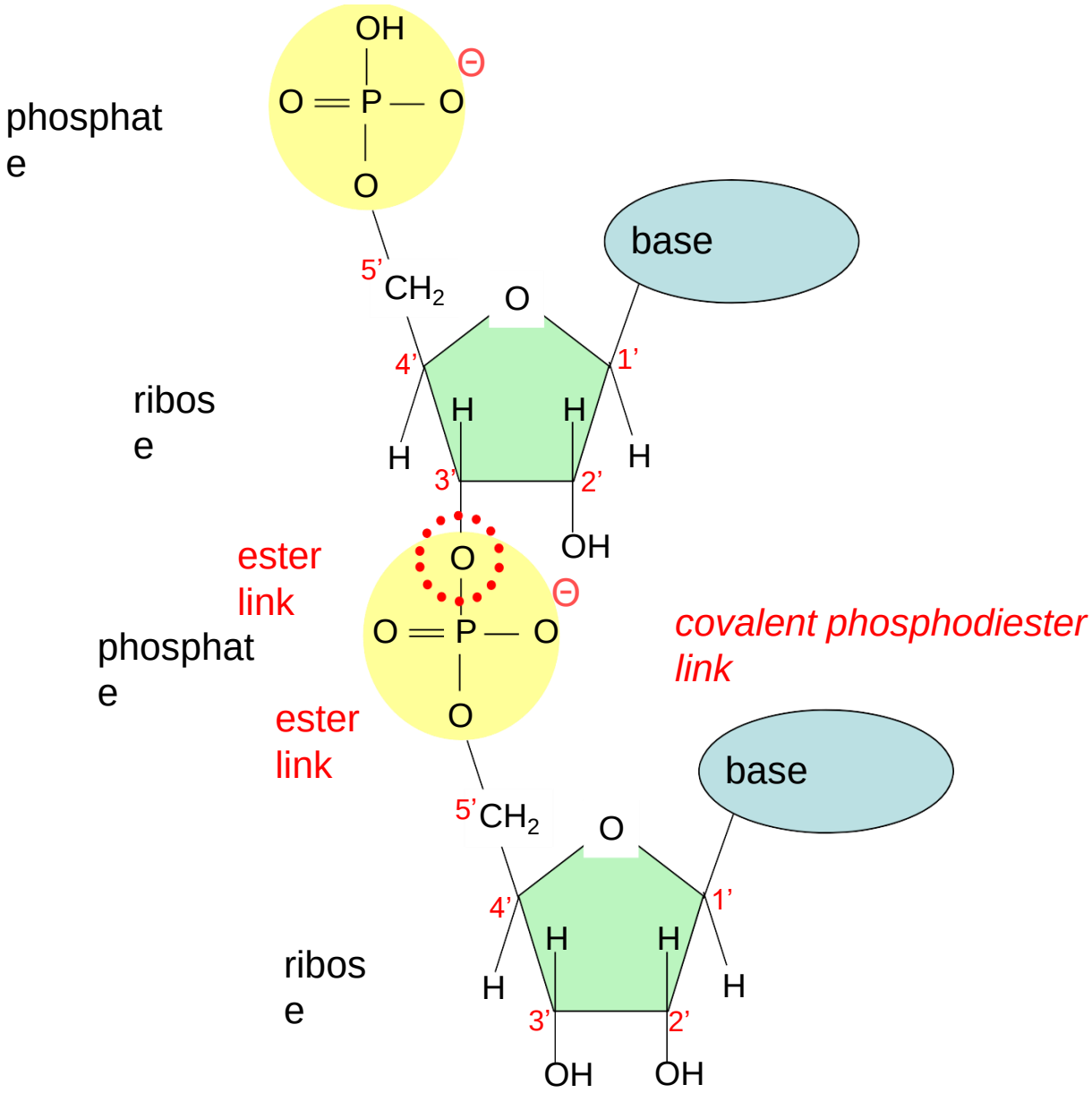


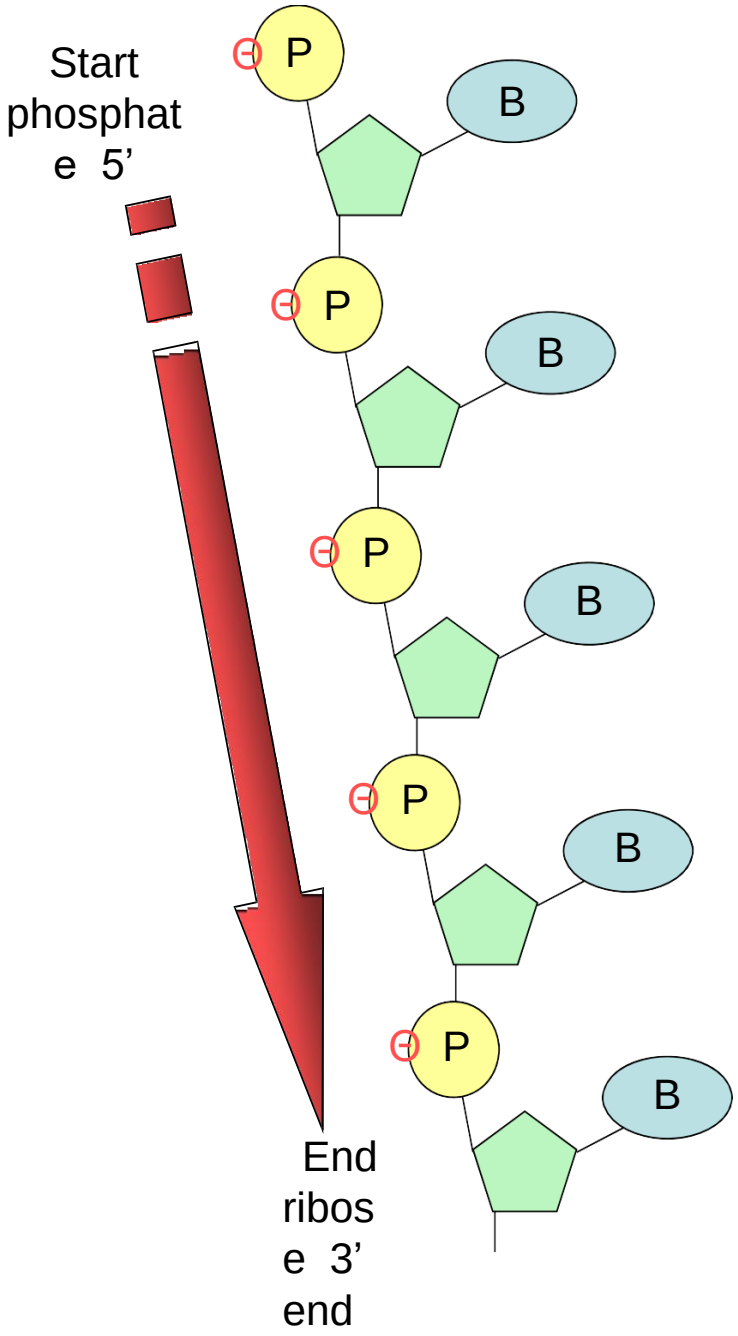
negative charge at neutral pH

sugar: ribose >> ribonucleic acid = RNA

or deoxyribose >> deoxyribonucleic acid = DNA


phosphat
e
OH
O
P
O
O
Θ
5’
CH2
base
O
4’
1’
H
H
H
H
3’
2’
OH
OH
ribos
e
condensation
OH
O
P
O
O
Θ
phosphat
e
5’
CH2
base
O
4’
1’
H
H
H
H
3’
2’
OH
OH
ribos
e

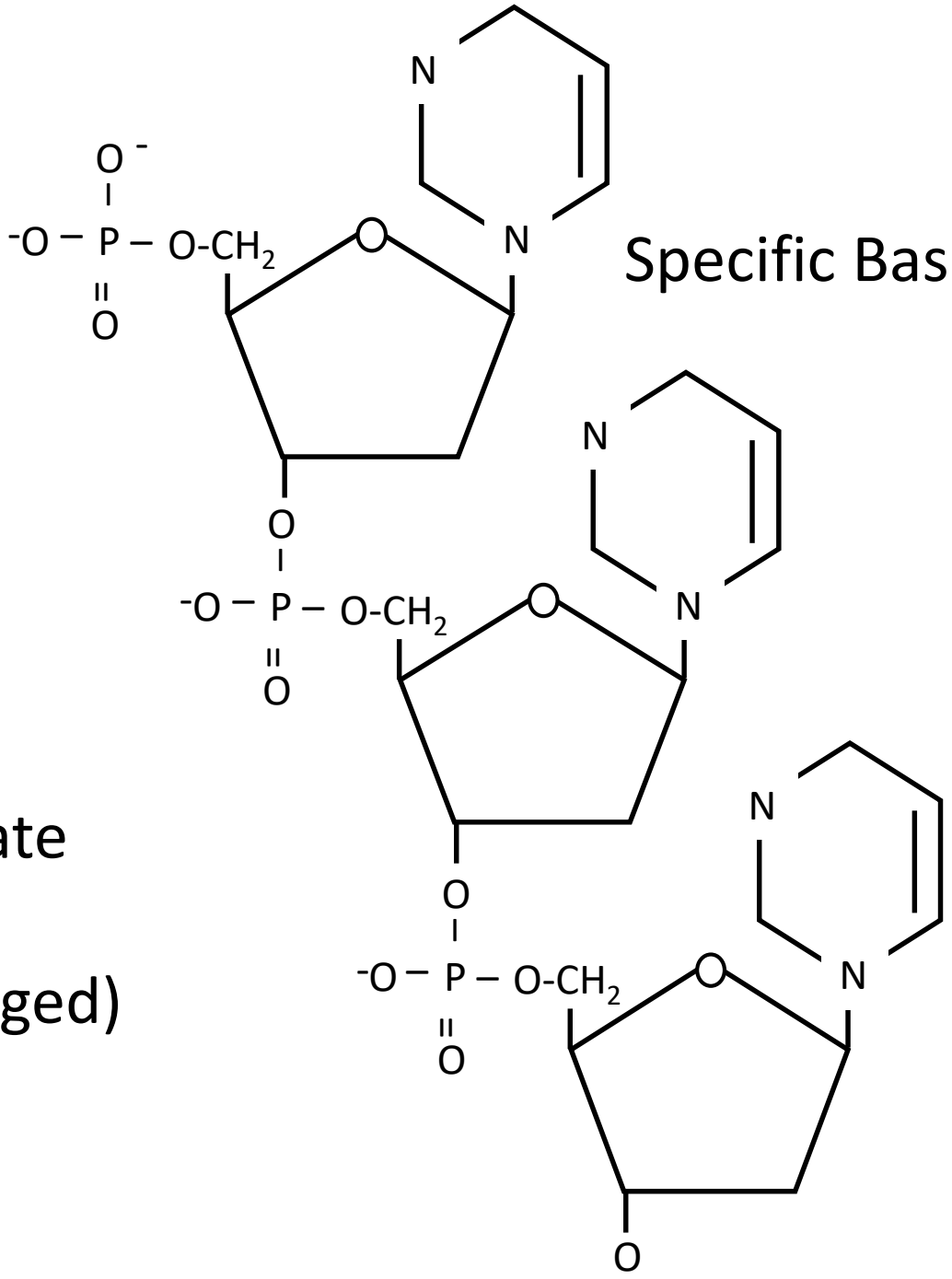

phosphat
e
OH
O
P
O
O
Θ
base
5’
CH2
O
4’
1’
H
H
H
H
3’
2’
OH
ribos
e
ester
link
O
O
P
O
O
Θ
covalent phosphodiester
link
phosphat
e
ester
link
base
5’
CH2
O
4’
1’
H
H
H
H
3’
2’
OH
OH
ribos
e



orientation :

synthesis direction
phosphate >>
ribose 5’ >> 3’

DNA



Sugar-Phosphate Backbone (negatively charged)

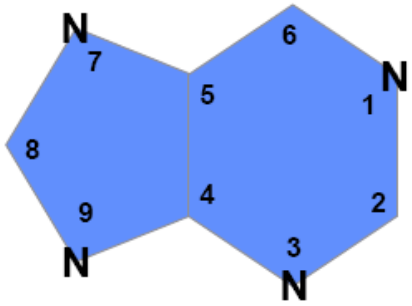
The Five Bases

- A = Adenine
- T = Thymine (only in DNA)
- G = Guanine
- C = Cytosine

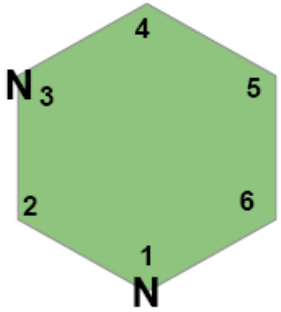
- RNA only:
  - U = Uracil (replaces T)

DNA review

Nitrogenous bases



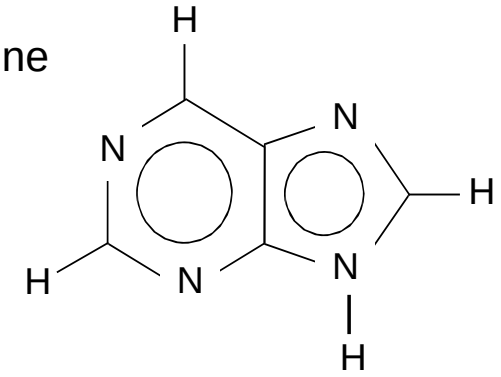
purine



pyrimidine

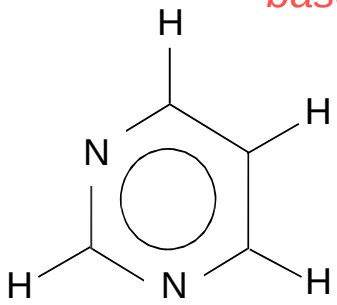
puric bases

Purine



pyrimidic bases

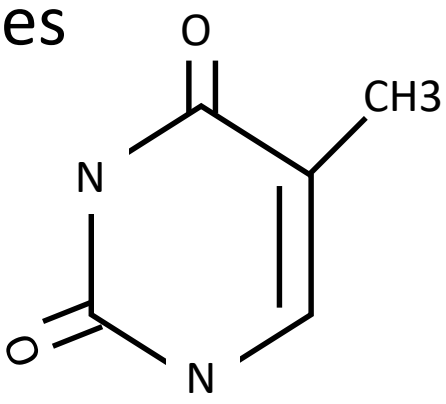
Pyrimidine



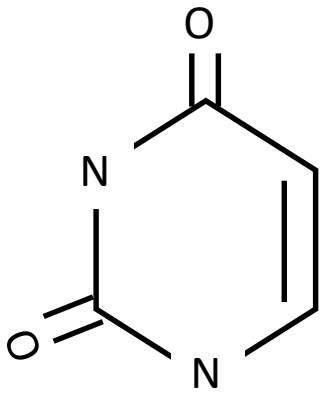
Cyclic aromatic plane structures Weak bases

Structures of Bases

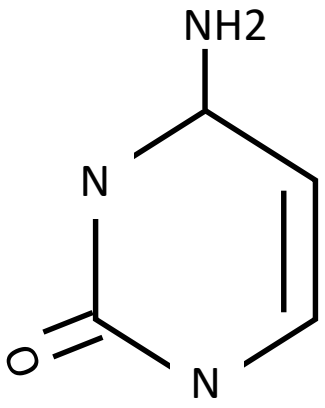
Pyrimidines



T

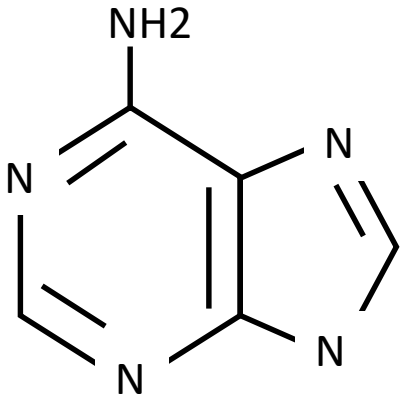


U

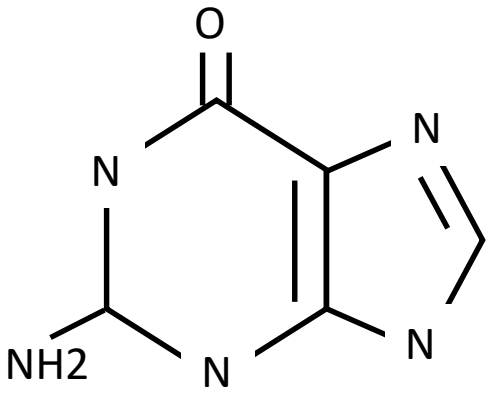


C

Purines



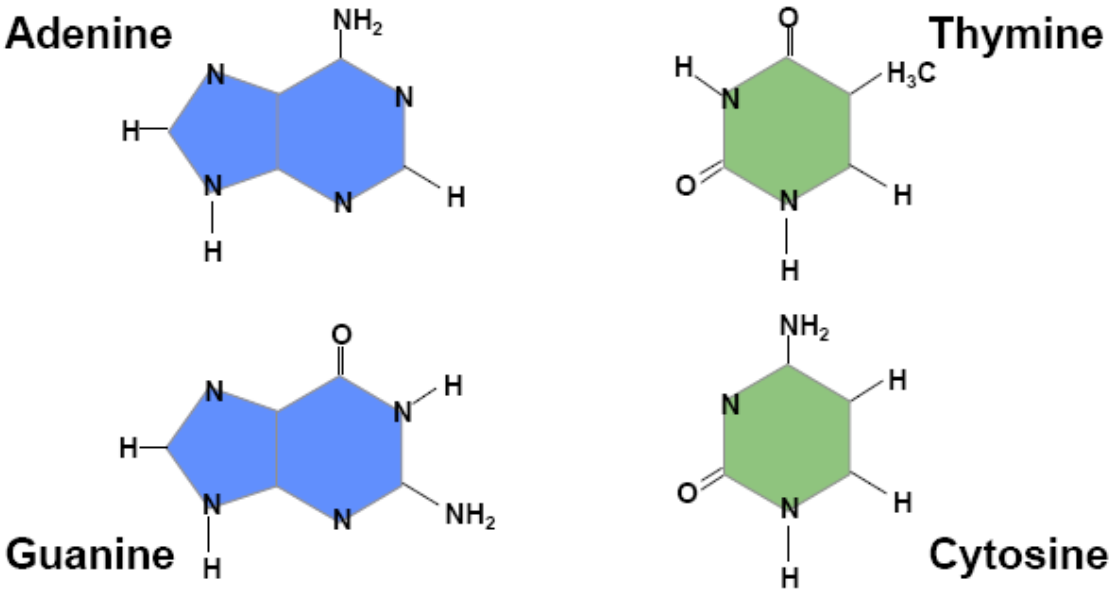
A



G

DNA review

Four bases in DNA

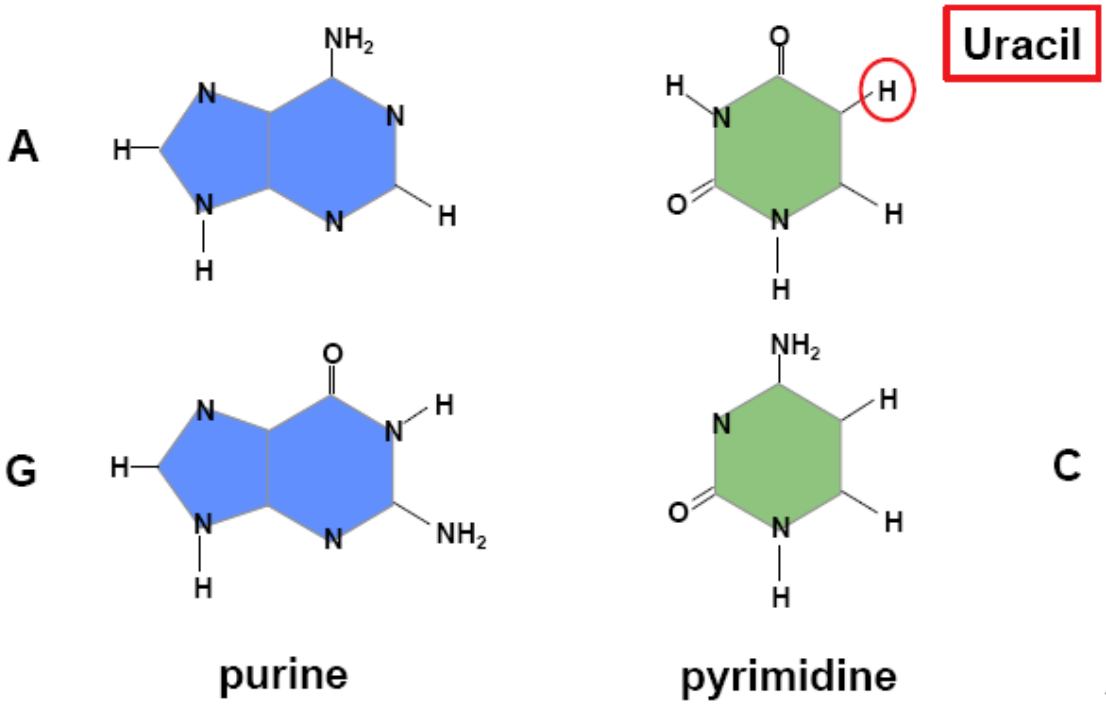


purine

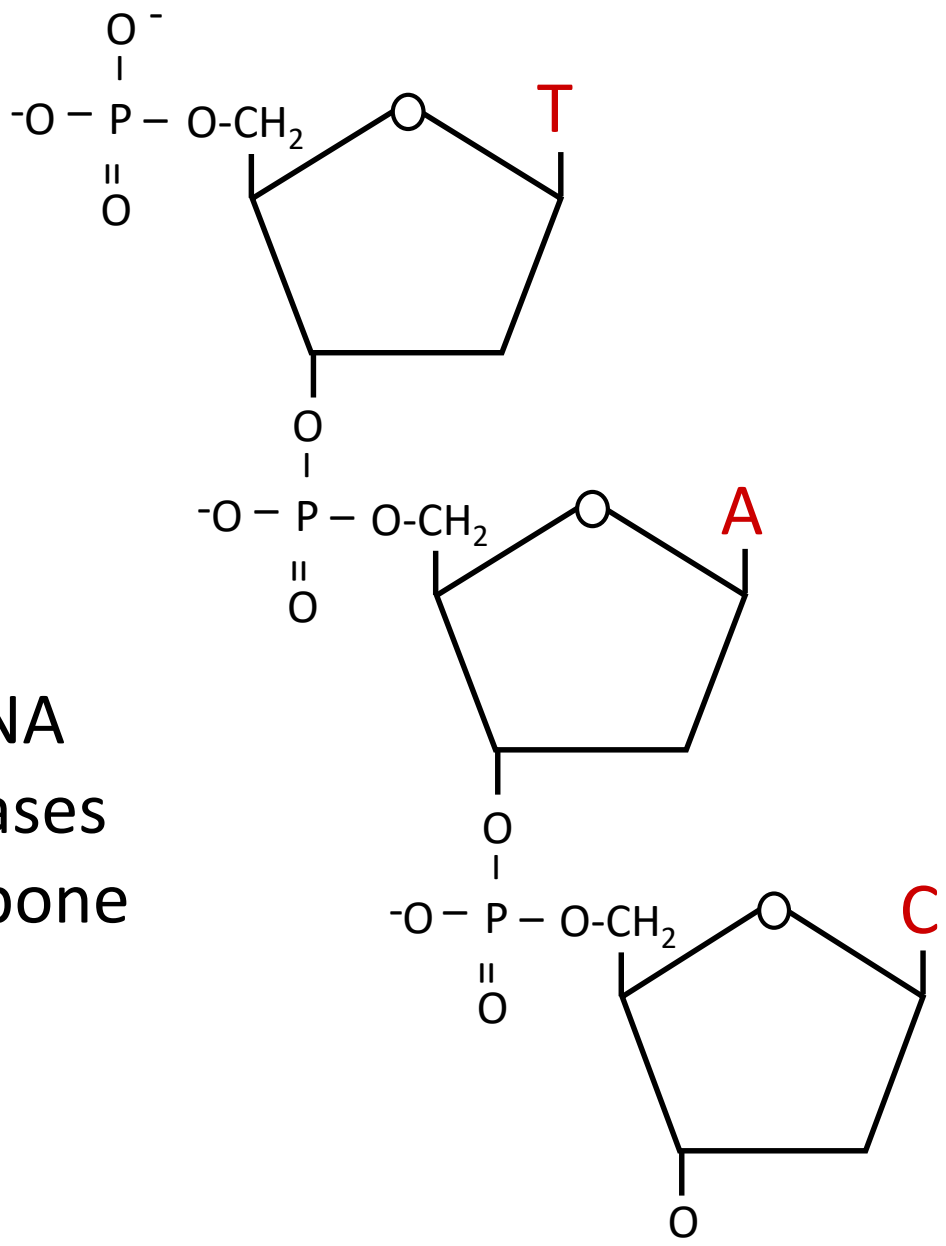
pyrimidine

DNA review

Thymine is replaced by Uracil in RNA



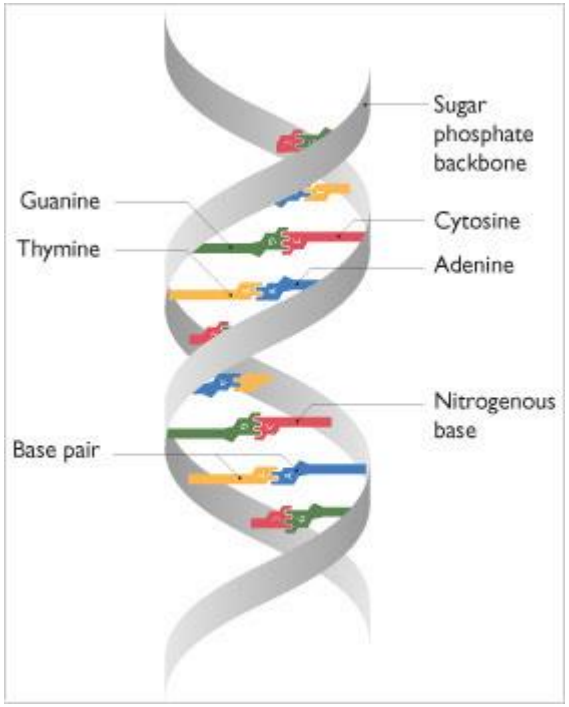
DNA



Sequence of DNA
is order of the bases
attached to backbone

Double Helix

- Sugar-Phosphate backbone is on outside
- Bases are inside - Hydrogen-bonding to opposing base on opposite strand

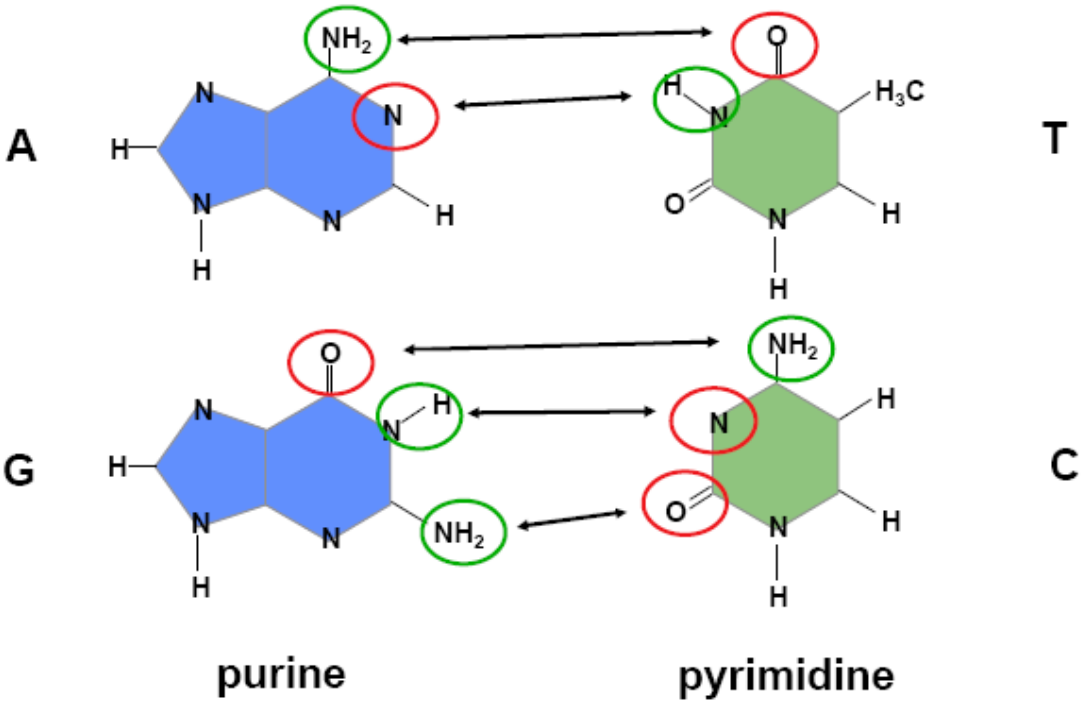


- Forming Base Pairs

DNA review

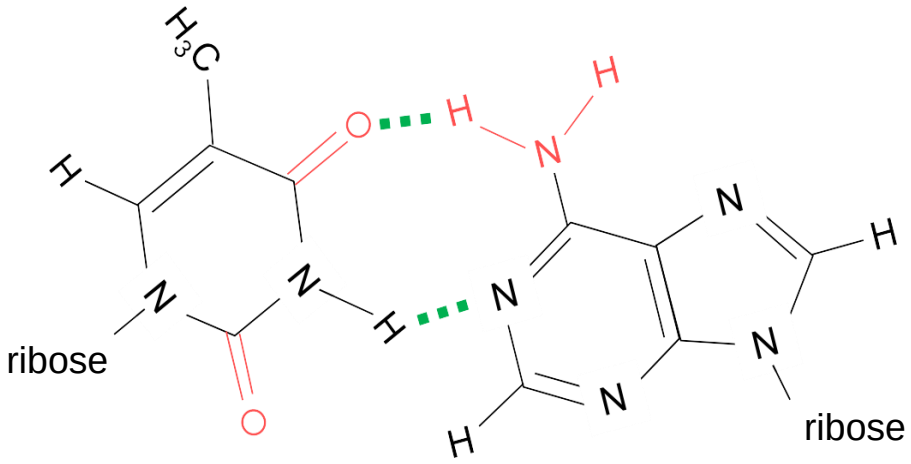
A-T base pairs have two hydrogen bonds

G-C base pairs have three hydrogen bonds



A=T pair

2 hydrogen bonds



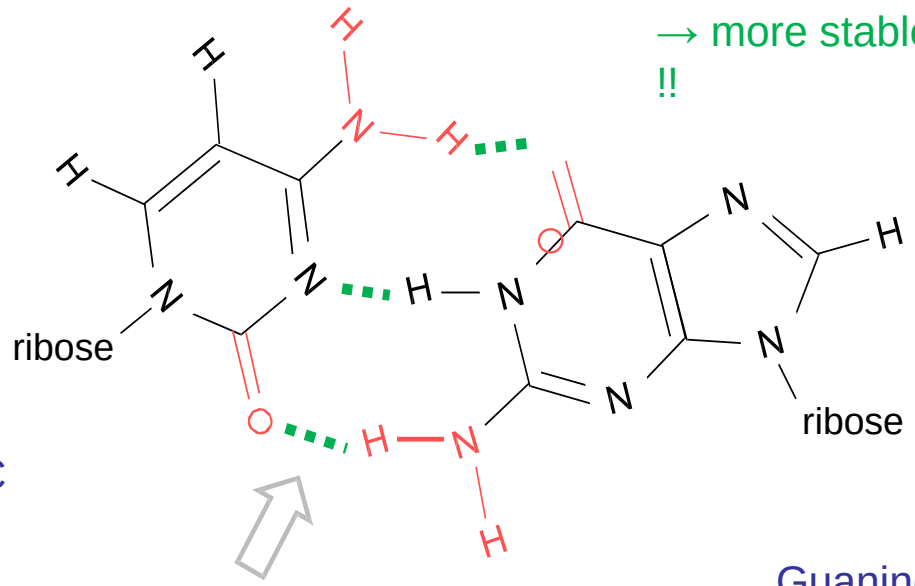
thymine T

adenine A

G≡C pair

3 hydrogen bonds

→ more stable !!



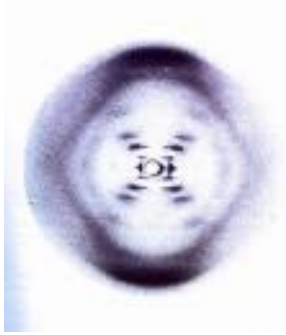
cytosine C

Guanine G

additional H-bond !

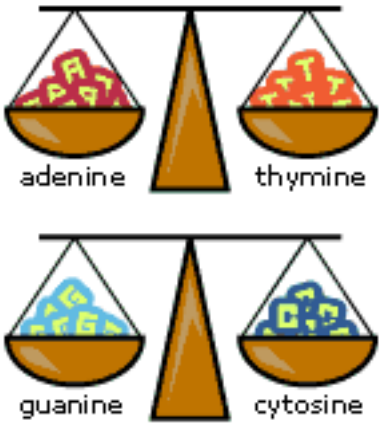
Discovering the structure of DNA

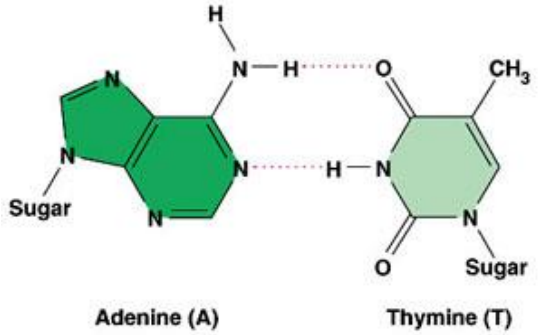
Rosalind Franklin's DNA image

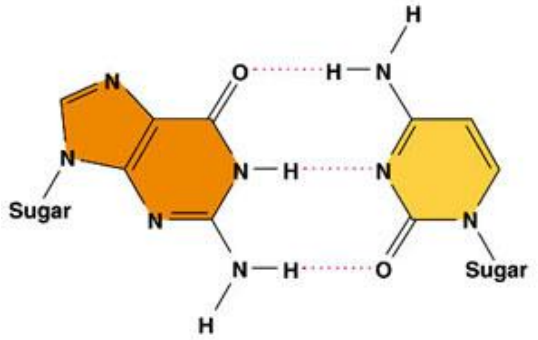


"Chargoff's rule"

A = T & C = G







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Base Pairing

1. A Purine must always be base paired to a Pyrimidine

2. A = T – with two Hydrogen Bonds

3. G ≡ C – with three Hydrogen Bonds

Therefore:

Strands must be complementary

Complementary Strands

Double Helix has two strands:

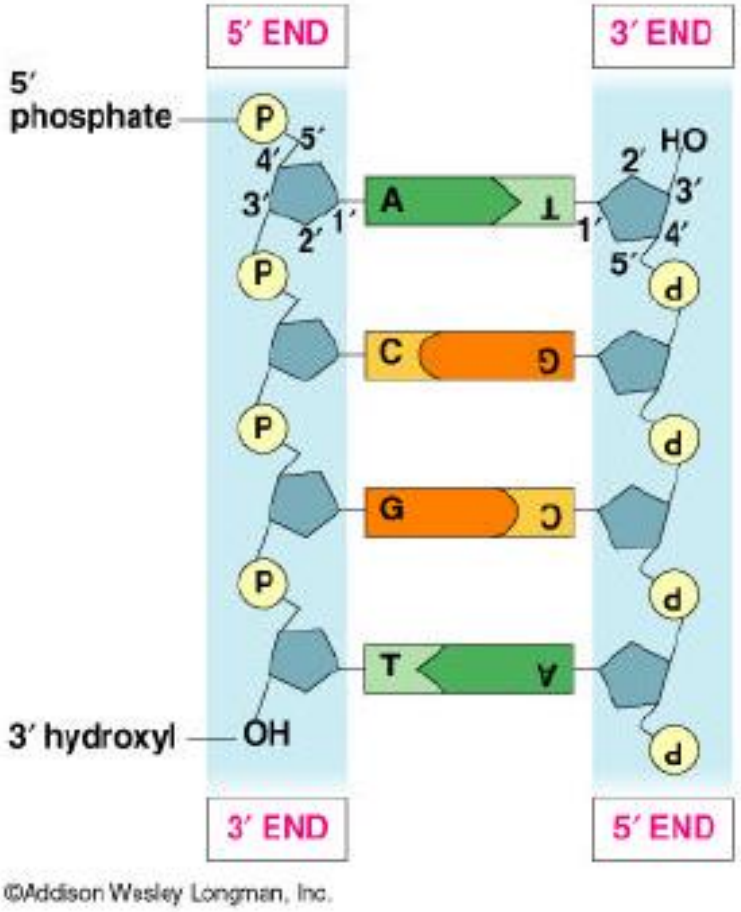
- Complementary – means when you read the message on one strand, you automatically know the message on other strand
- Not identical, because in reverse
- “Antiparallel” strands
- Exact same message on both strands

Antiparallel Strands

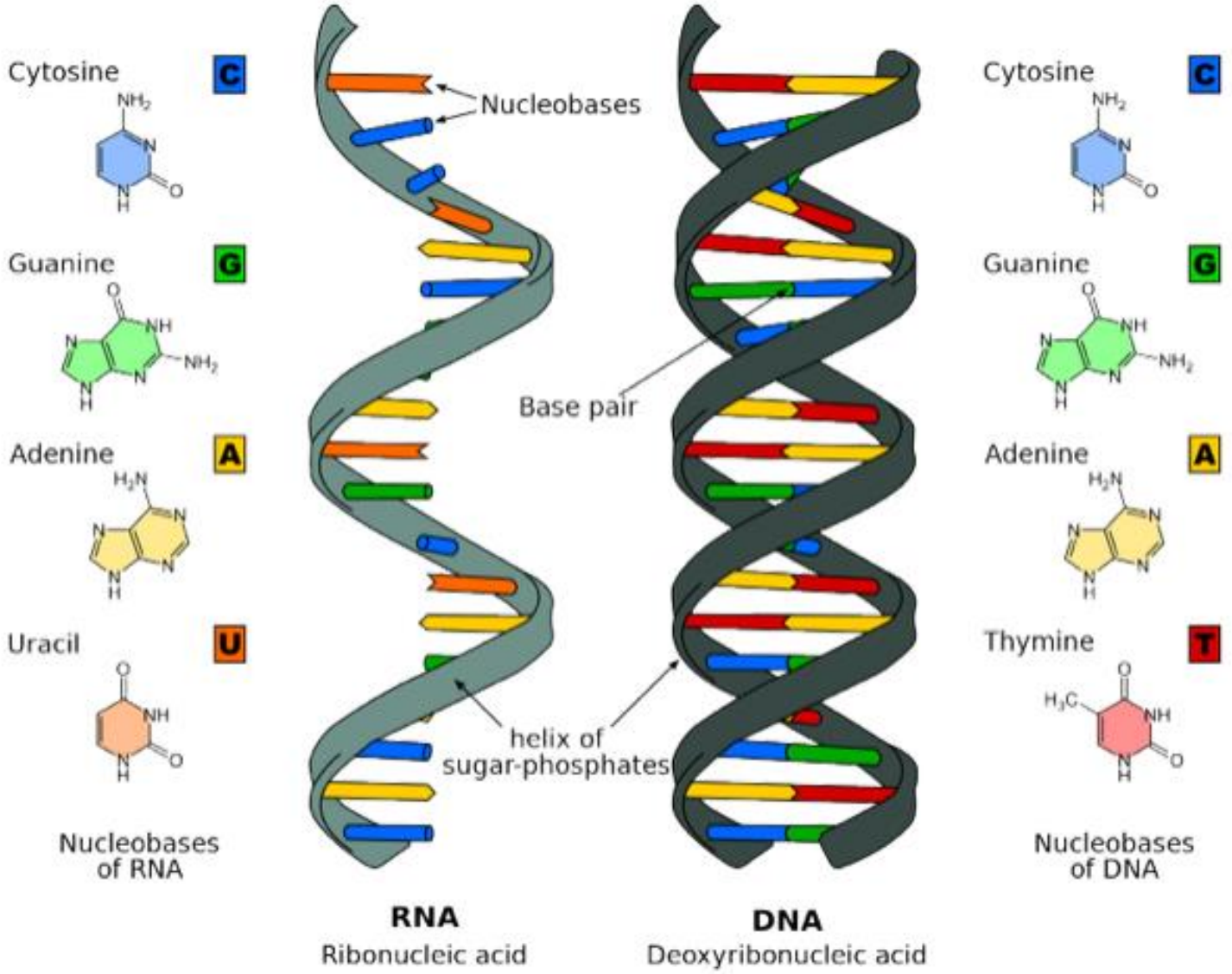
- DNA strands match up in opposite directions
- DNA always “read”

  5’ to 3’ direction

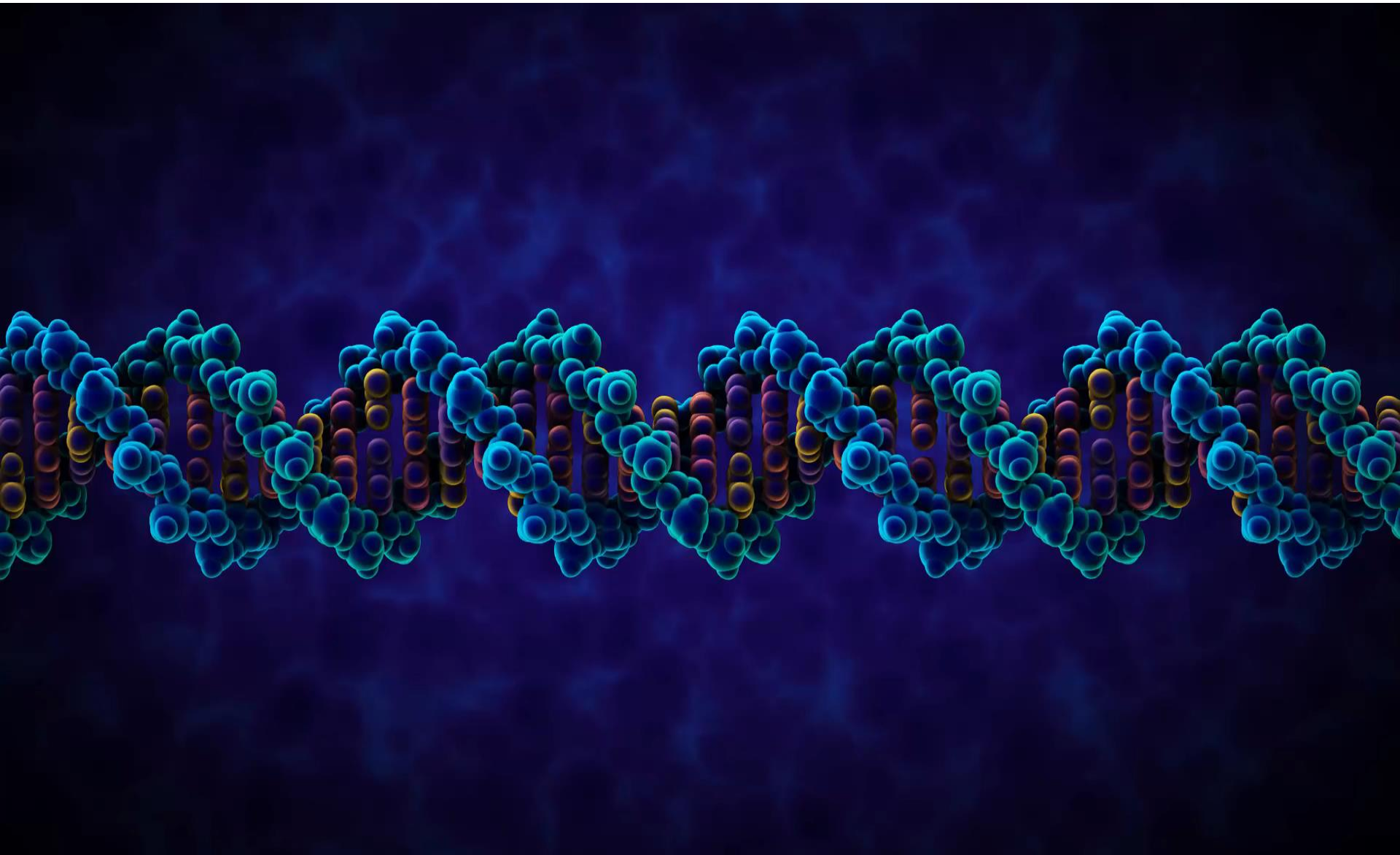
- In the end, both strands have the exact same message



DNA Structure and RNA structure:



Watch clip

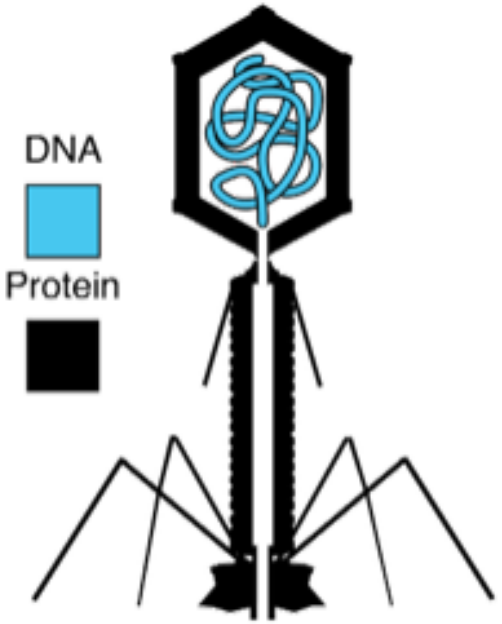


Differences between DNA and RNA

	DNA	RNA
Sugar	deoxyribose	ribose
Bases	A, G, C, T	A, G, C, U
Strands	Double strands	Single strand
Genetic material	Most life	Some viruses

Viral genomes

- Genome can be:
    1. single stranded DNA.
    2. double stranded DNA.
    3. single stranded RNA.
    4. double stranded RNA.



- Genome can be linear or circular.

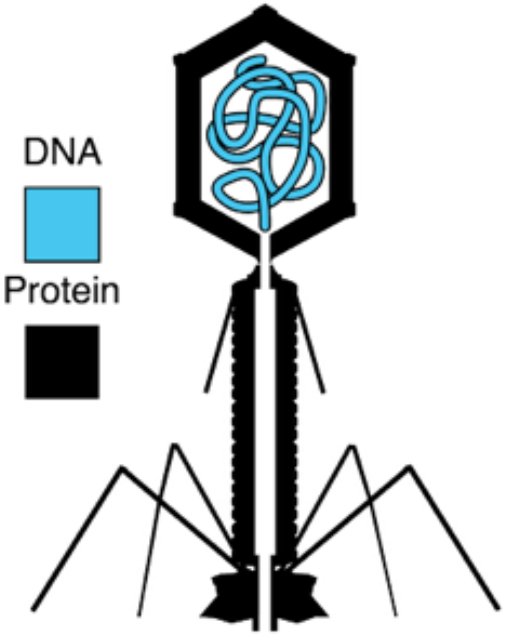
Viral genomes

•One molecule or segmented genome (many pieces).

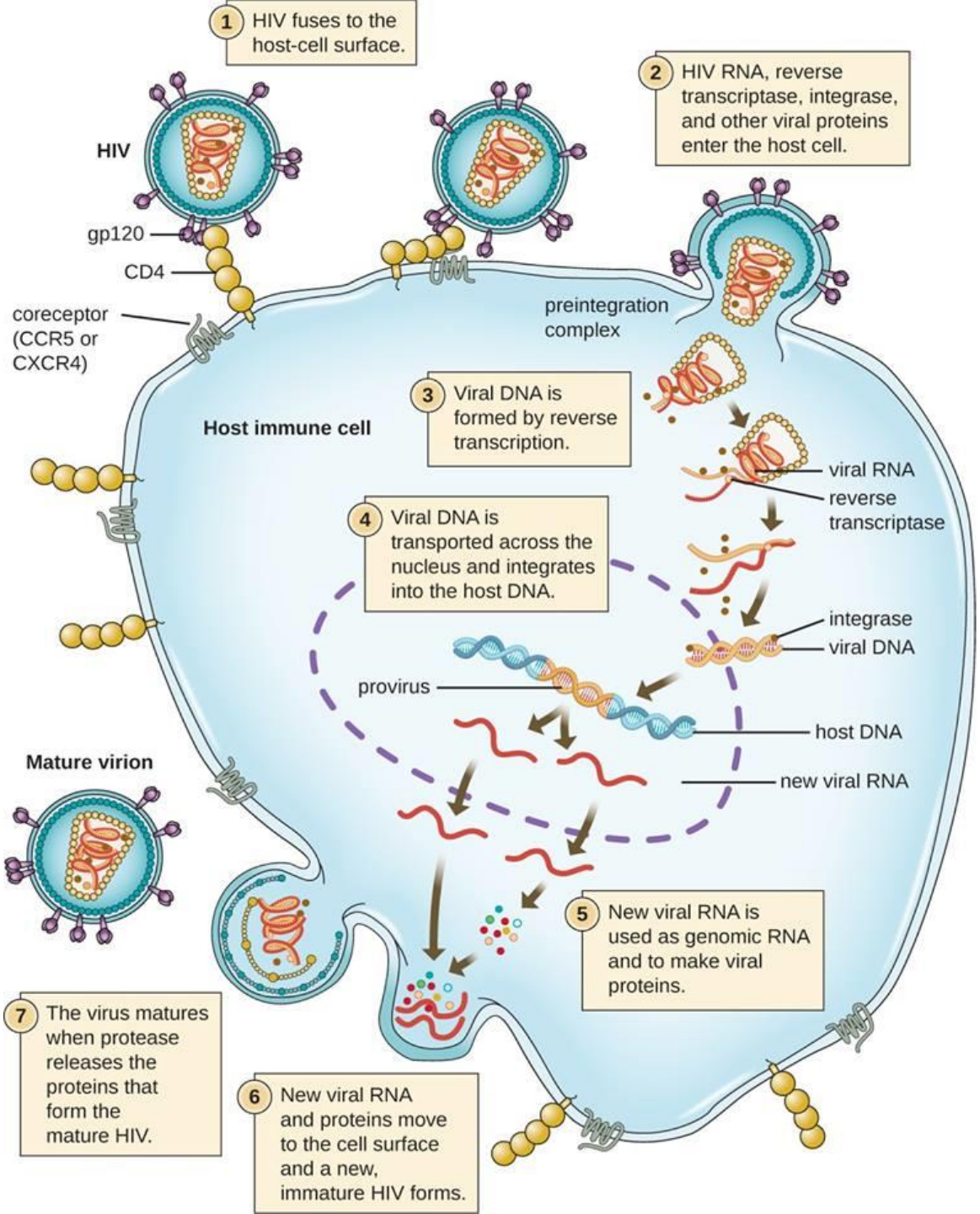
• Viruses with RNA genome are called retroviruses!

• Genome size 2 thousand basepairs (2Kb) – 2 million basepairs (2 Mb).

• No special organization of the genome.



The HIV lifecycle



Prokaryotic genomes

- Most genomes are:

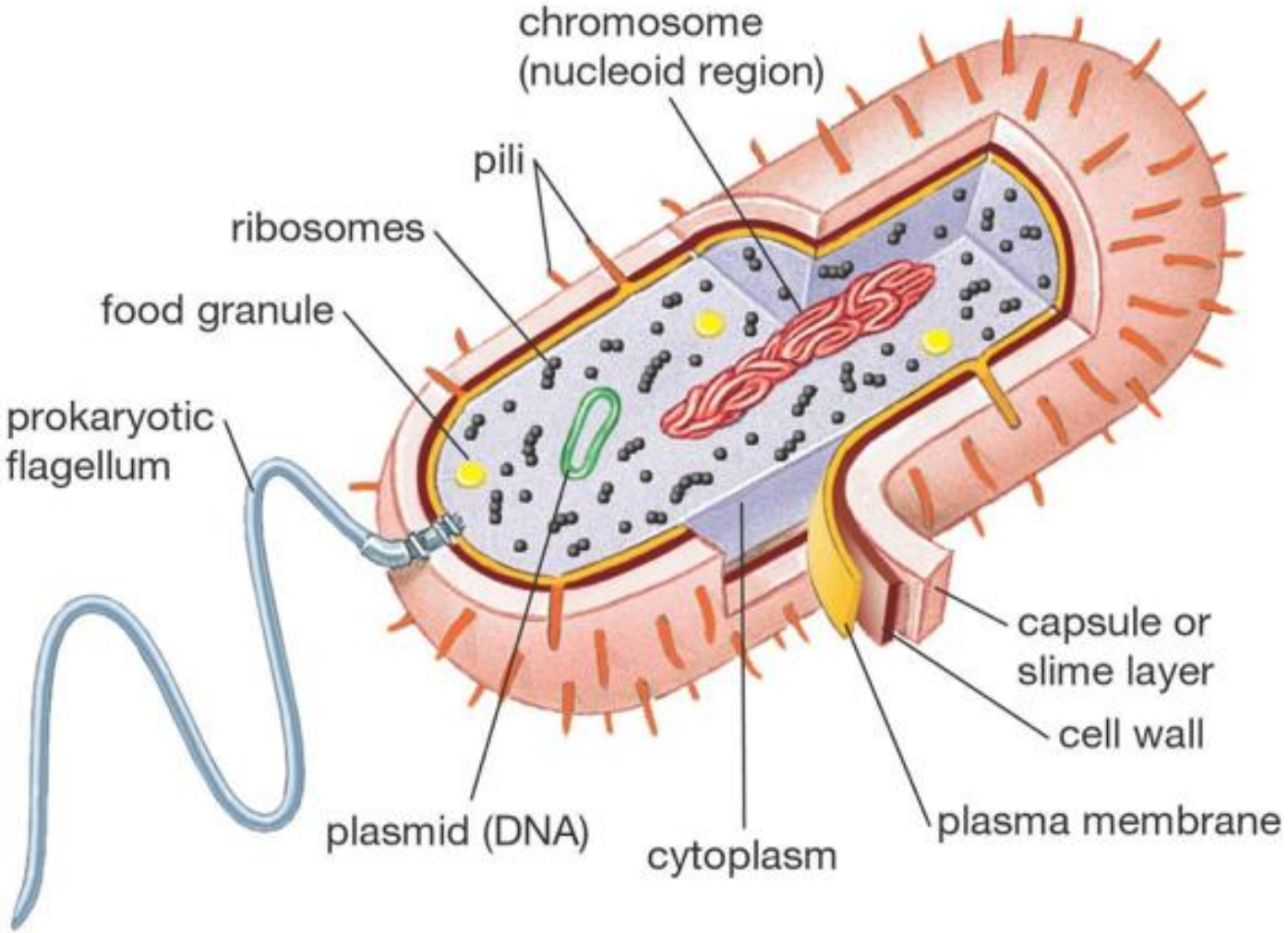
1. Single molecule
2. One circular chromosome.
3. Double strands DNA.
4. Some prokaryotic cells have small additional circular DNA that can replicate independently (plasmid).

- Genome is organised in a structure called nucleoid.

Prokaryotic genomes

- The genome is loose and not surrounded by a membrane (different than eukaryotes).
- **Loop domain** in bacterial genome contribute to the packaging.
- To fit the genome in a bacterial cell the DNA undergoes **supercoiling**.

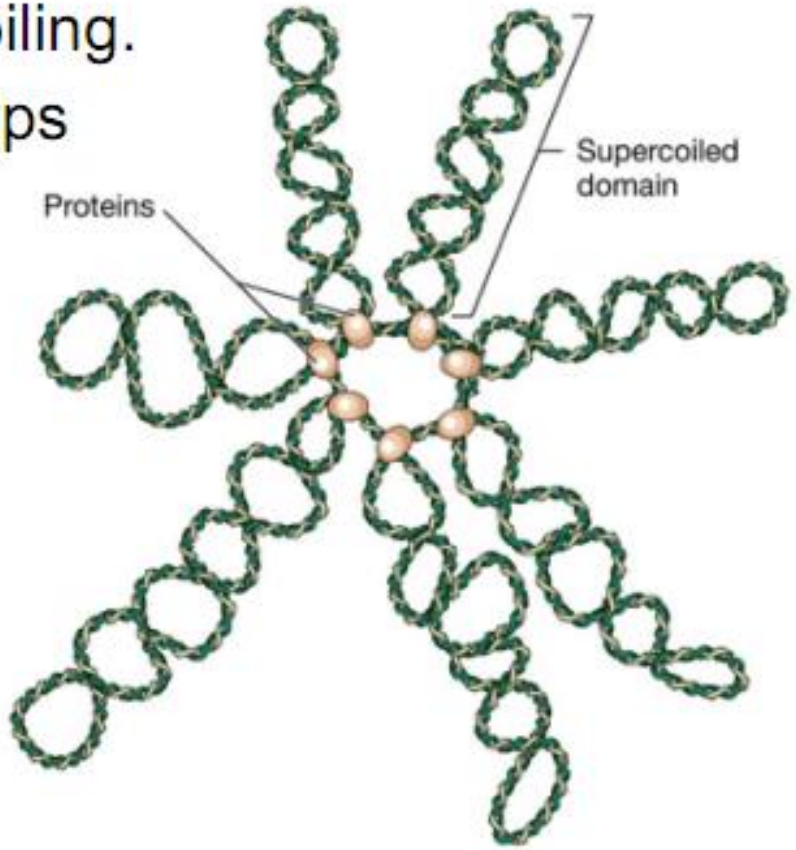
Prokaryotic genomes



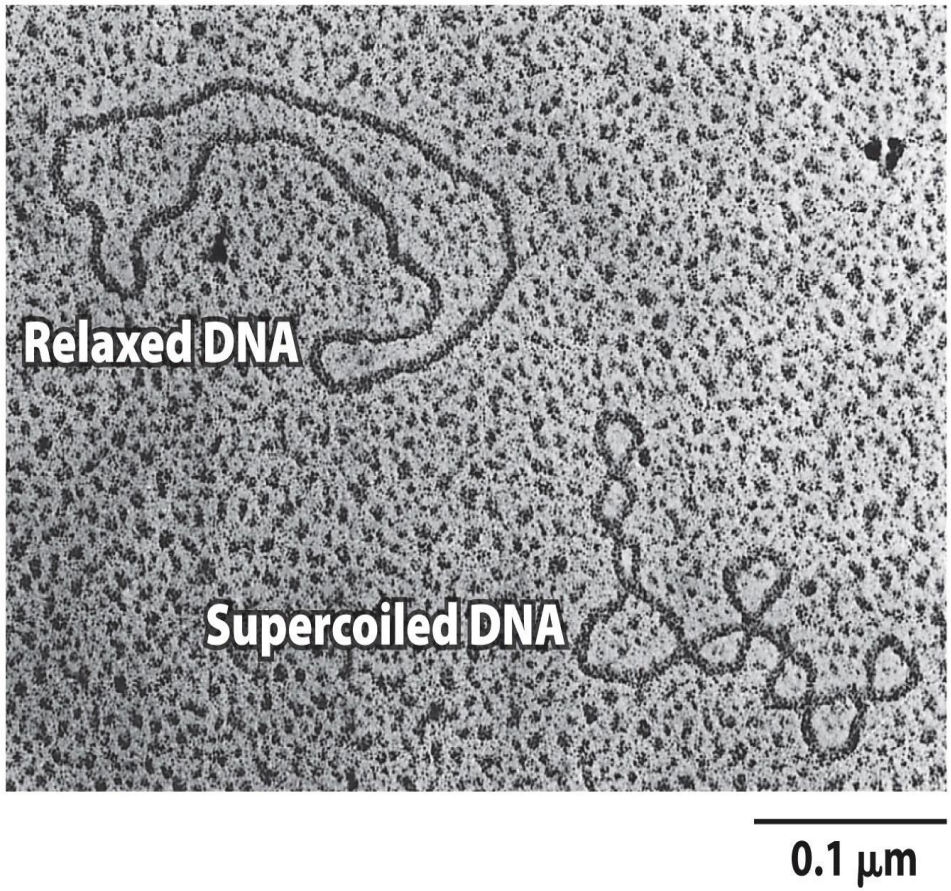
Prokaryotic genomes organization

Prokaryote genome organization through:

1. Nucleoid (protein DNA interaction).
2. Supercoiling.
3. DNA loops



DNA Structure: Supercoils



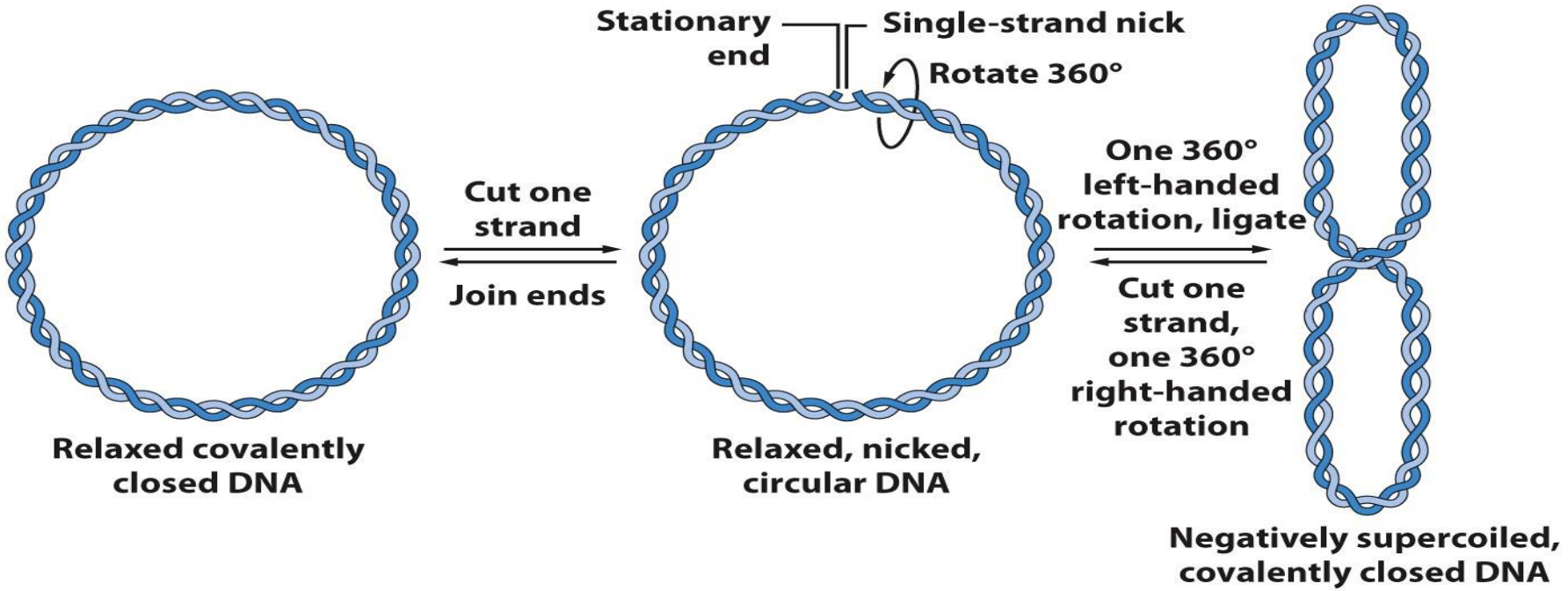
Courtesy Dr. Svend Freytag

Unique structure of a DNA molecule

Is produced after one or both strand of DNA are cleaved

DNA will rotate or twist

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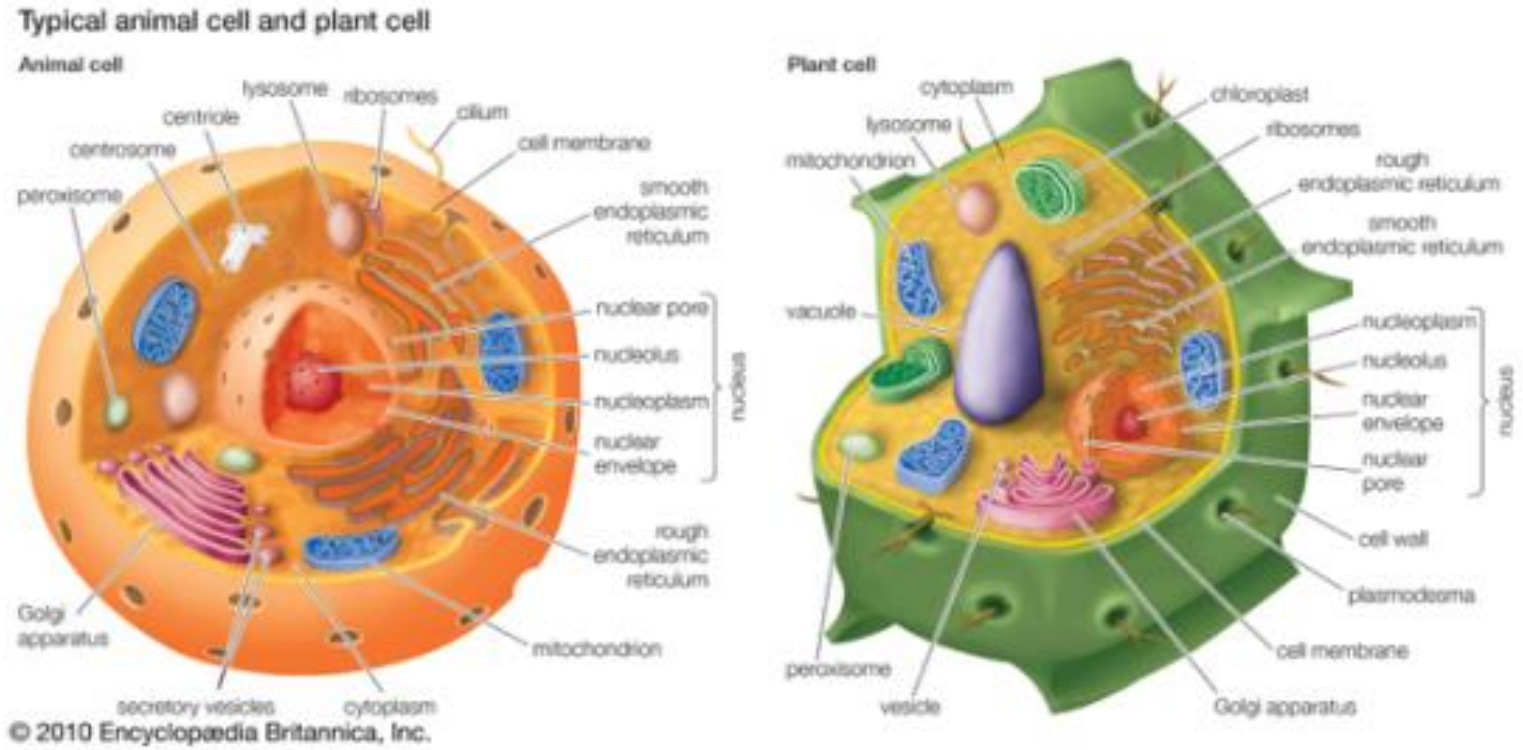


ns



Eukaryotic genomes

Where is eukaryotic genome located?



Nuclear genome, mitochondrial genome, chloroplast genome

Eukaryotic genomes

1. Eukaryotic genome contains double strands DNA.

2. Eukaryotic genome is arranged in several linear packages called chromosomes through interaction with protein Histone.

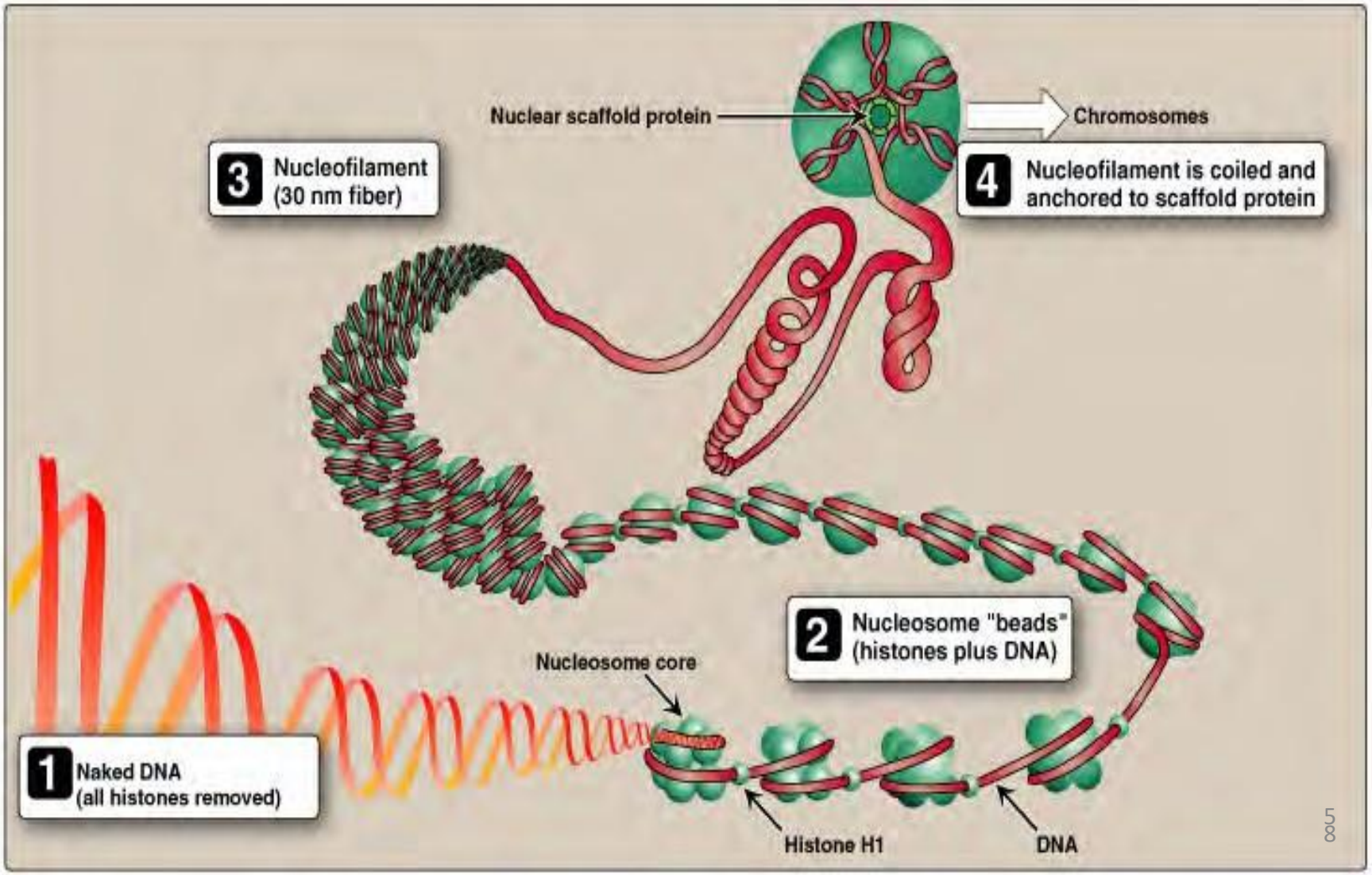
Chromosome Structure in Eukaryotes

Eukaryotic chromosomes contain huge molecules of DNA that are **highly condensed** during mitosis and meiosis.

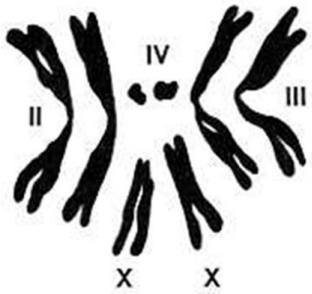
The centromeres and telomeres of eukaryotic chromosomes have **unique structures**.

Each chromosome contains a single large double helix (strand) of DNA molecule

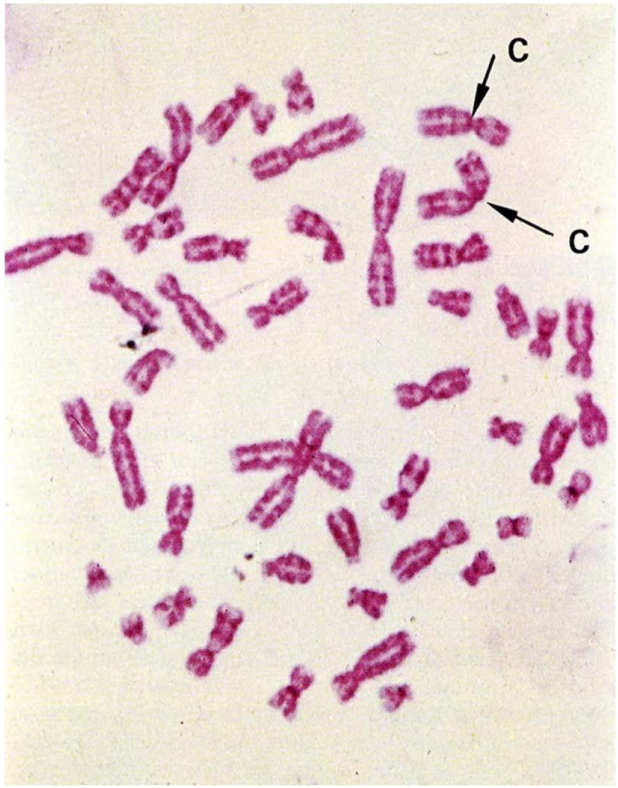
DNA is packed into chromosomes during mitosis and meisosis



chromosomes can be seen by microscopy

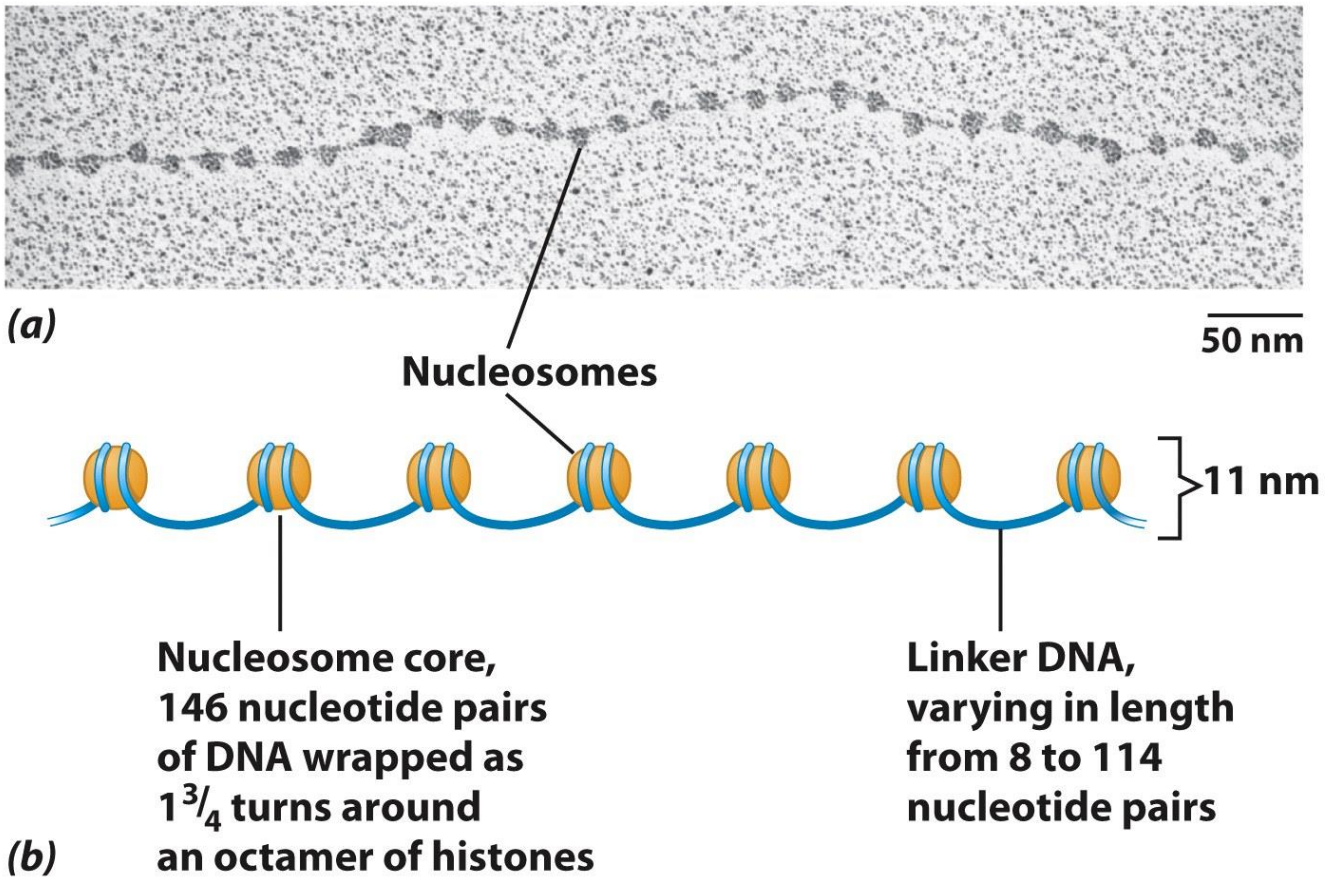


fruit fly 4 chromosome pairs



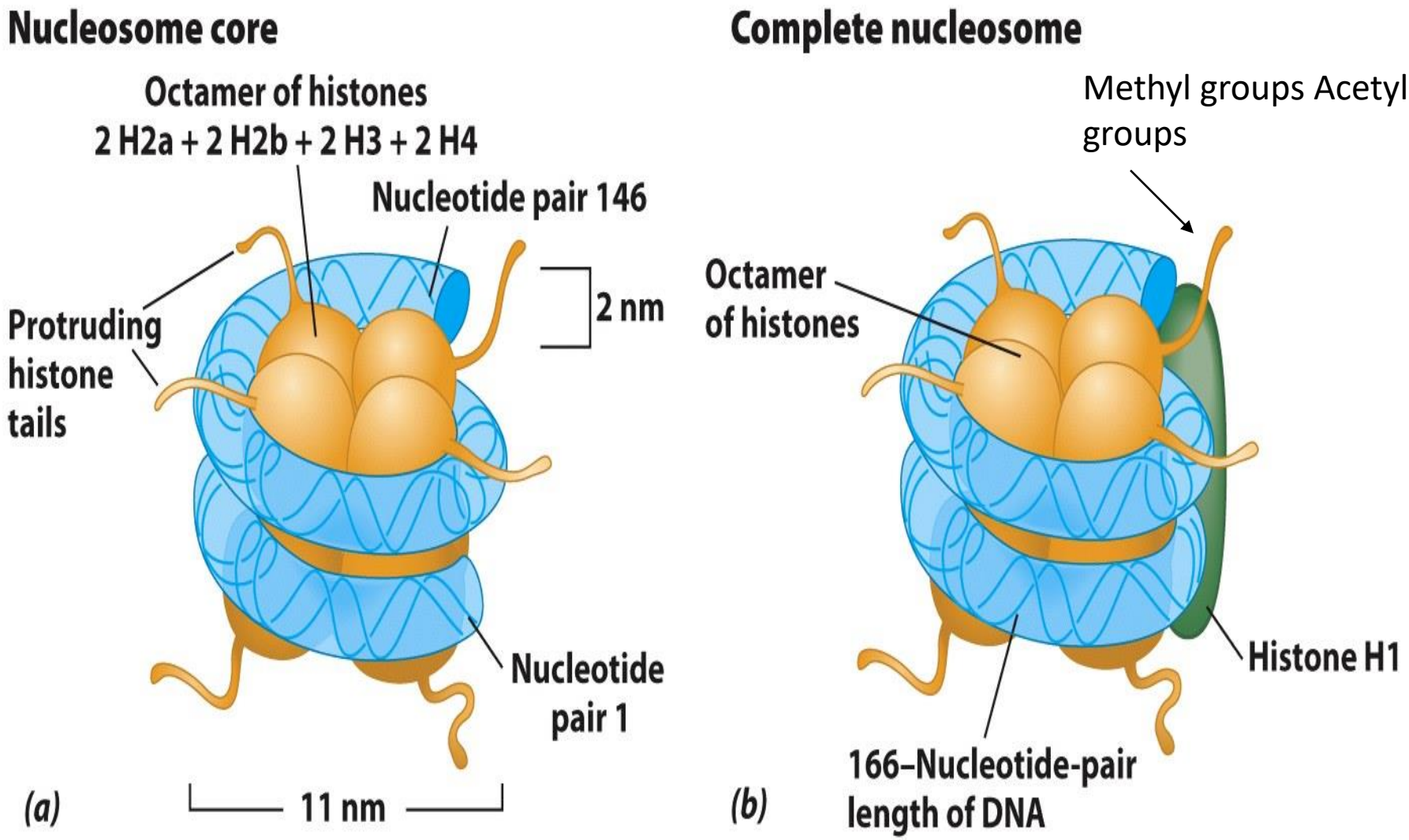
human 23 chromosome pairs

Nucleosomes



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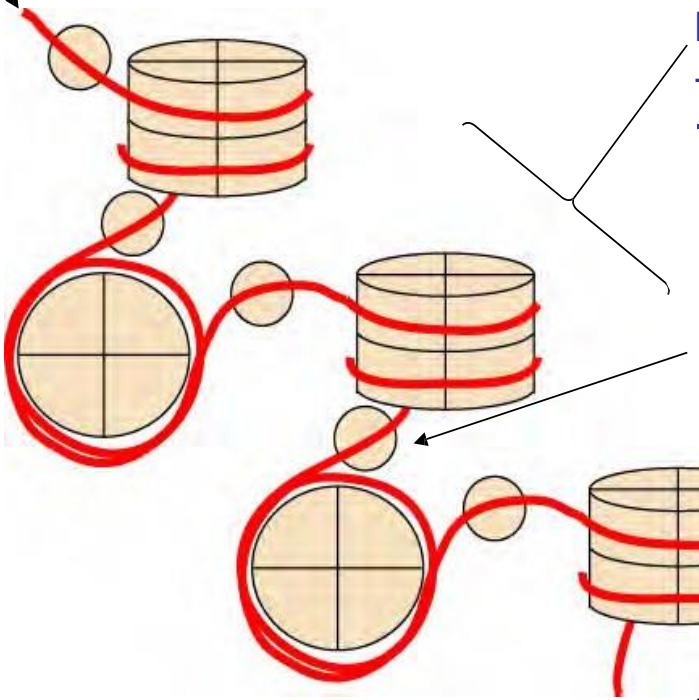


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Chromatin

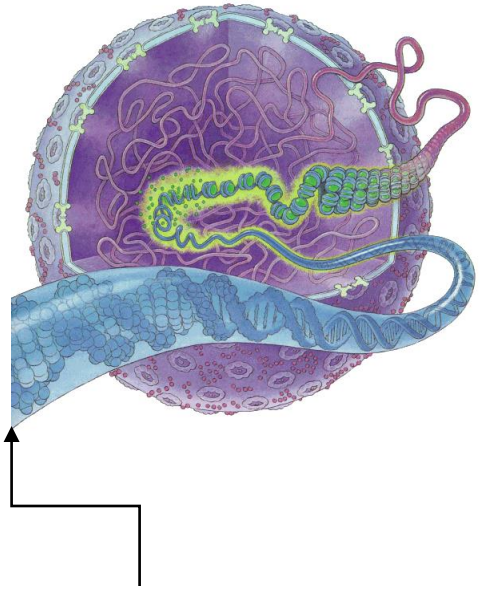
In eukaryotes, DNA is associated with proteins called *histones*



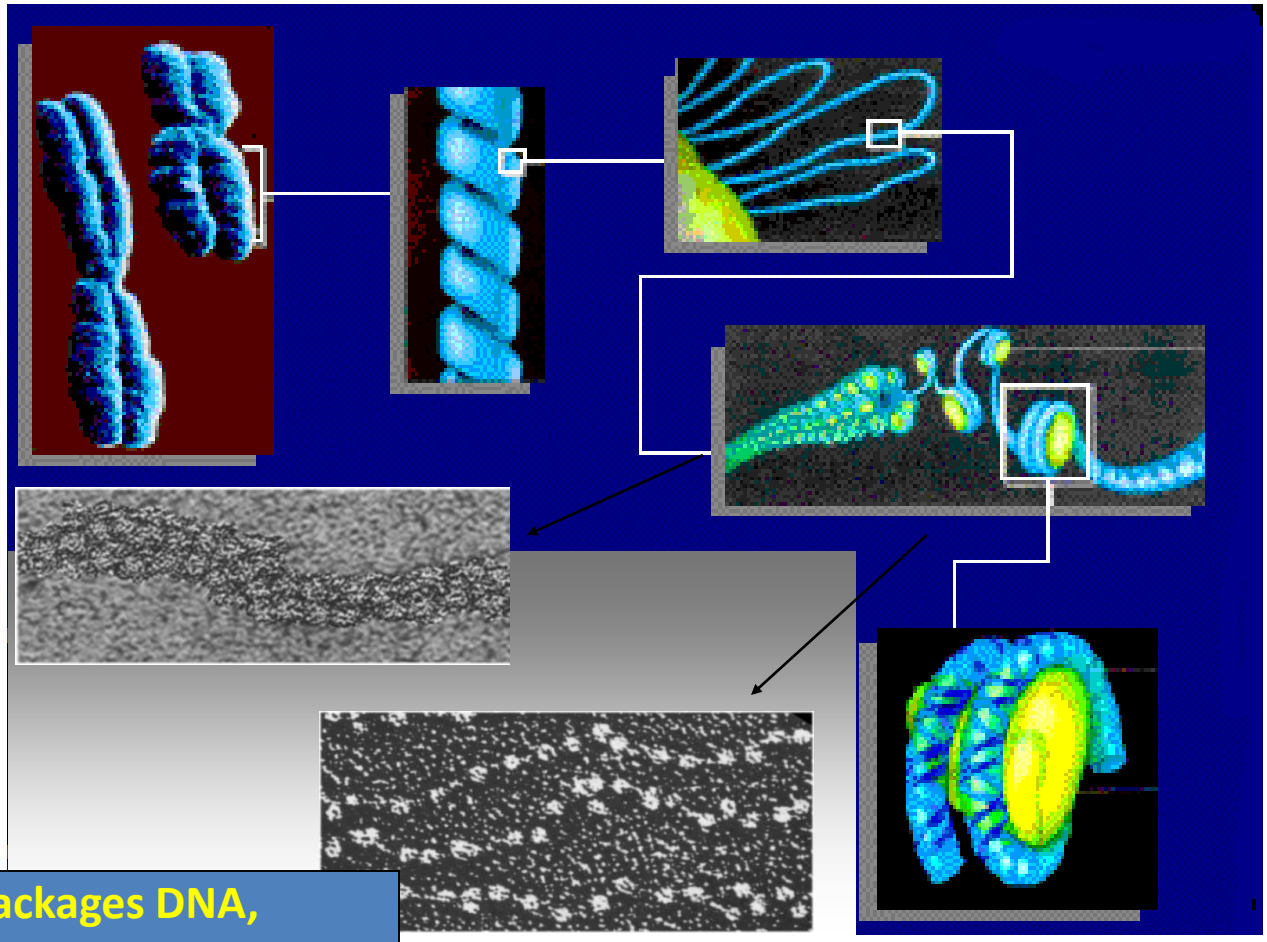
Nucleosome:
- DNA (1,65 turn)
- 2 x 4 histones : H2A
  H2B
  H3 H4
- Histone H1

Chromatin fiber Ø ±30 nm

Packing of DNA into chromatin



2 m of DNA in a nucleus with a diameter of 5-10 μm



Chromatin not only packages DNA, but also regulates DNA accessibility through modifications in chromatin structure.

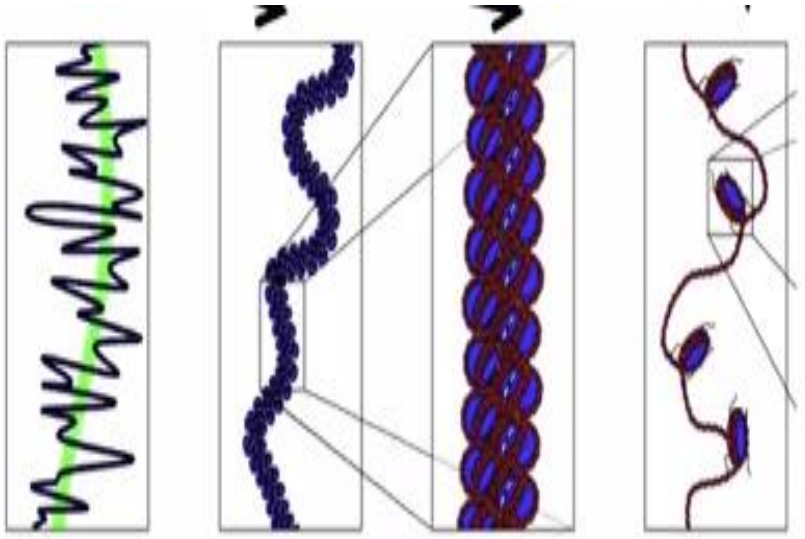
Heterochromatin & Euchromatin

Heterochromatin: closed chromatin, densely stained

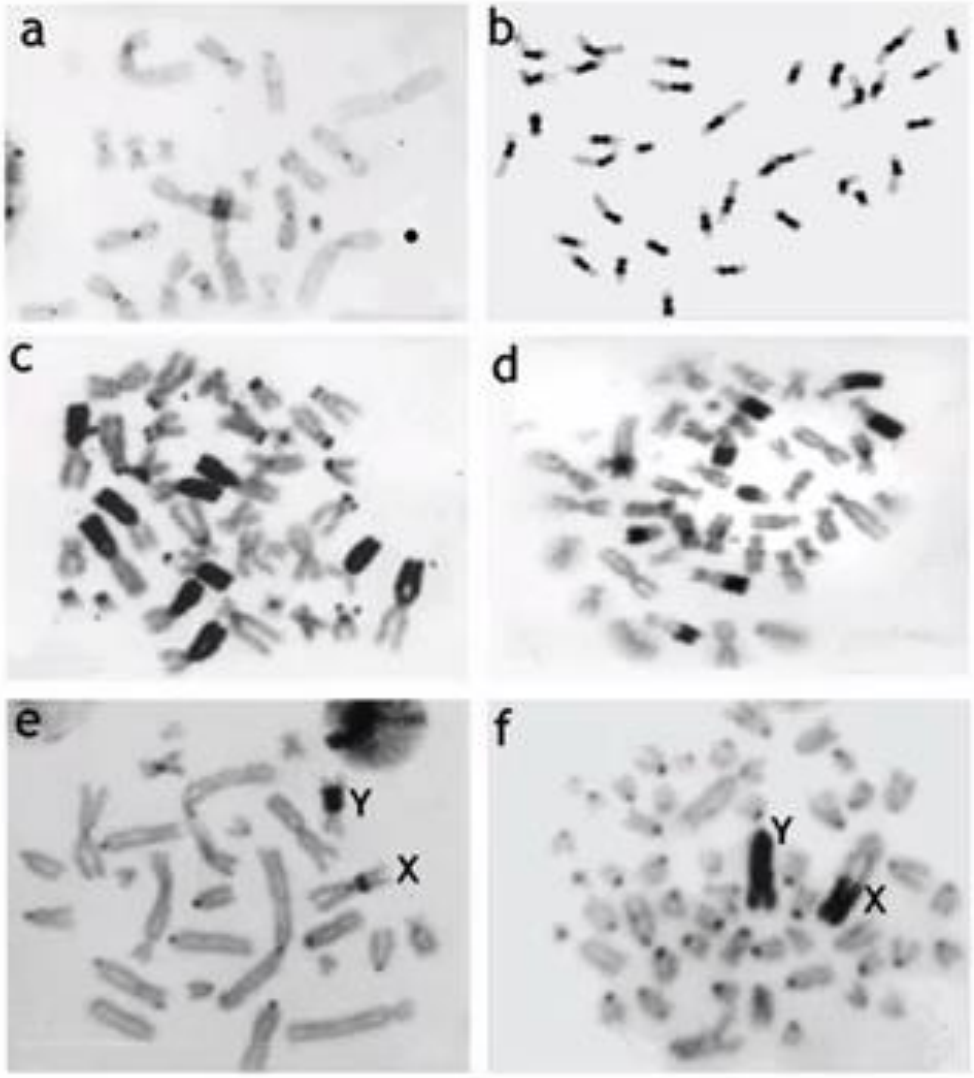
Euchromatin: opened chromatin, less stained

- Falcultative – can differ by cell type or time.
  Ex: Tissue specific genes.
  Inactive X chromosome.

- Constitutive – same in all cell types. Performs a structural role.
  Ex: Centromeres.
  Telomeres.
  Portions of sex chromosome (mainly Y)



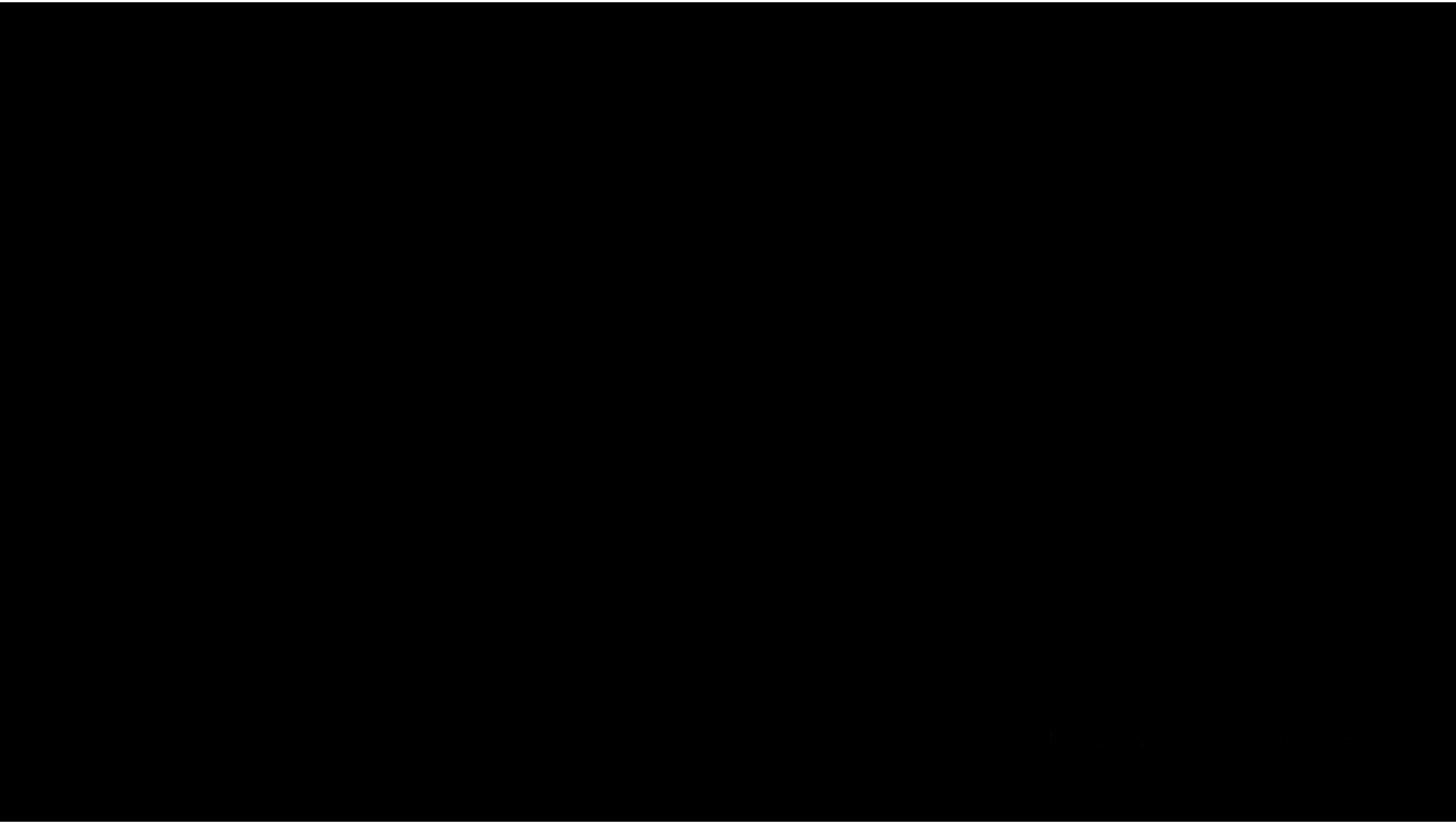
Constitutive heterochromatin

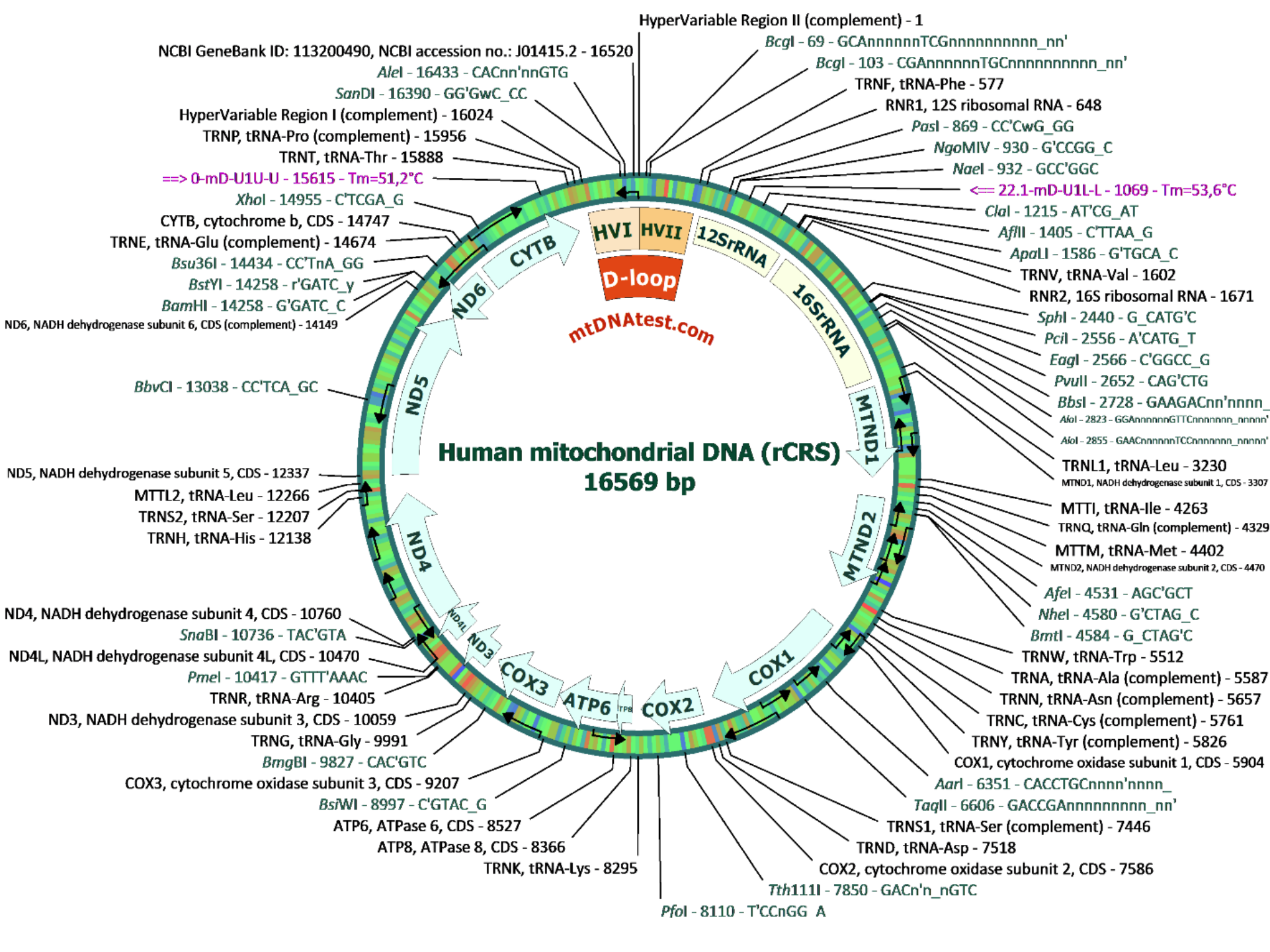


Metaphase chromosomes from different mammals show different amounts of constitutive heterochromatin at:

1. Centromeres (a & b)
2. Telomeres (d)
3. Autosomal chromosome arms (c)
4. Sex chromosomes (e and f)

Endosymbiotic theory




HyperVariable Region II (complement) - 1
BcgI - 69 - GCAnnnnnnTCGnnnnnnnnnn_nn'
BcgI - 103 - CGAnnnnnnTGCnnnnnnnnnn_nn'
TRNF, tRNA-Phe - 577
RNR1, 12S ribosomal RNA - 648
PasI - 869 - CC'CwG_GG
NgoMIV - 930 - G'CCGG_C
NaeI - 932 - GCC'GGC
<== 22.1-mD-U1L-L - 1069 - Tm=53,6°C
ClaI - 1215 - AT'CG_AT
AflII - 1405 - C'TTAA_G
ApaLI - 1586 - G'TGCA_C
TRNV, tRNA-Val - 1602
RNR2, 16S ribosomal RNA - 1671
SphI - 2440 - G_CATG'C
PciI - 2556 - A'CATG_T
EagI - 2566 - C'GGCC_G
PvuII - 2652 - CAG'CTG
BbsI - 2728 - GAAGACnn'nnnn_
AloI - 2823 - GGAnnnnnnGTTCnnnnnnn_nnnnn'
AloI - 2855 - GAACnnnnnnTCCnnnnnnn_nnnnn'
TRNL1, tRNA-Leu - 3230
MTND1, NADH dehydrogenase subunit 1, CDS - 3307
MTTI, tRNA-Ile - 4263
TRNQ, tRNA-Gln (complement) - 4329
MTTM, tRNA-Met - 4402
MTND2, NADH dehydrogenase subunit 2, CDS - 4470
AfeI - 4531 - AGC'GCT
NheI - 4580 - G'CTAG_C
BmtI - 4584 - G_CTAG'C
TRNW, tRNA-Trp - 5512
TRNA, tRNA-Ala (complement) - 5587
TRNN, tRNA-Asn (complement) - 5657
TRNC, tRNA-Cys (complement) - 5761
TRNY, tRNA-Tyr (complement) - 5826
COX1, cytochrome oxidase subunit 1, CDS - 5904
AarI - 6351 - CACCTGCnnnn'nnnn_
TaqII - 6606 - GACCGAnnnnnnnnn_nn'
TRNS1, tRNA-Ser (complement) - 7446
TRND, tRNA-Asp - 7518
COX2, cytochrome oxidase subunit 2, CDS - 7586
Tth111I - 7850 - GACn'n_nGTC
PfoI - 8110 - T'CCnGG_A
TRNK, tRNA-Lys - 8295
ATP8, ATPase 8, CDS - 8366
ATP6, ATPase 6, CDS - 8527
BsiWI - 8997 - C'GTAC_G
COX3, cytochrome oxidase subunit 3, CDS - 9207
BmgBI - 9827 - CAC'GTC
TRNG, tRNA-Gly - 9991
ND3, NADH dehydrogenase subunit 3, CDS - 10059
TRNR, tRNA-Arg - 10405
PmeI - 10417 - GTTT'AAAC
ND4L, NADH dehydrogenase subunit 4L, CDS - 10470
SnaBI - 10736 - TAC'GTA
ND4, NADH dehydrogenase subunit 4, CDS - 10760
TRNH, tRNA-His - 12138
TRNS2, tRNA-Ser - 12207
MTTL2, tRNA-Leu - 12266
ND5, NADH dehydrogenase subunit 5, CDS - 12337
BbvCI - 13038 - CC'TCA_GC
ND6, NADH dehydrogenase subunit 6, CDS (complement) - 14149
BamHI - 14258 - G'GATC_C
BstYI - 14258 - r'GATC_y
Bsu36I - 14434 - CC'TnA_GG
TRNE, tRNA-Glu (complement) - 14674
CYTB, cytochrome b, CDS - 14747
XhoI - 14955 - C'TCGA_G
==> 0-mD-U1U-U - 15615 - Tm=51,2°C
TRNT, tRNA-Thr - 15888
TRNP, tRNA-Pro (complement) - 15956
HyperVariable Region I (complement) - 16024
SanDI - 16390 - GG'GwC_CC
AleI - 16433 - CACnn'nnGTG
NCBI GeneBank ID: 113200490, NCBI accession no.: J01415.2 - 16520
HVI
HVII
D-loop
12SrRNA
16SrRNA
MTND1
MTND2
COX1
COX2
ATP6
ATP8
COX3
ND3
ND4L
ND4
ND5
ND6
CYTB
mtDNAtest.com
Human mitochondrial DNA (rCRS)
16569 bp

Human mitochondrial genome

Table 1 | **Comparison between the human nuclear and mitochondrial genomes***

Characteristic	Nuclear genome	Mitochondrial genome
Size	~3.3 x 10^9 bp	16,569 bp
Number of DNA molecules per cell	23 in haploid cells; 46 in diploid cells	Several thousand copies per cell (polyploidy)
Number of genes encoded	~20,000–30,000	37 (13 polypeptides, 22 tRNAs and 2 rRNAs)
Gene density	~1 per 40,000 bp	1 per 450 bp
Introns	Frequently found in most genes	Absent
Percentage of coding DNA	~3%	~93%
Codon usage	The universal genetic code	AUA codes for methionine; TGA codes for tryptophan; AGA and AGG specify stop codons
Associated proteins	Nucleosome-associated histone proteins and non-histone proteins	No histones; but associated with several proteins (for example, TFAM) that form nucleoids
Mode of inheritance	Mendelian inheritance for autosomes and the X chromosome; paternal inheritance for the Y chromosome	Exclusively maternal
Replication	Strand-coupled mechanism that uses DNA polymerases α and δ	Strand-coupled and strand-displacement models; only uses DNA polymerase γ
Transcription	Most genes are transcribed individually	All genes on both strands are transcribed as large polycistrons
Recombination	Each pair of homologues recombines during the prophase of meiosis	There is evidence that recombination occurs at a cellular level but little evidence that it occurs at a population level

Summary of key concepts

- DNA is the chemical basis of heredity.
- In DNA, base pairing (A=T ; C≡G) creates a structure that is helical, double stranded and anti-parallel. This structure is lost in case of DNA denaturation.
- The DNA sequence is read from 5' to 3'.
- DNA and RNA have different chemical properties that reflect their different biological functions.
- In eukaryotes, DNA is packed around proteins (histones), forming a complex called chromatin.
- Further compaction of chromatin at some stages of the cell cycle leads to the formation of chromosomes, which can be observed using a microscope.